🤖 AGENTIC DESIGN PATTERNS

A Complete Learning Curriculum: Beginner to Advanced

📋 TABLE OF CONTENTS

EXECUTIVE SUMMARY

What are Agentic Design Patterns?

Agentic design patterns are reusable architectural blueprints for building AI systems that can autonomously make decisions, take actions, and learn from interactions. They define HOW an AI system (agent) perceives its environment, reasons about problems, and executes solutions with minimal human intervention.

Why Now?

- LLMs like GPT-4, Claude, and others have moved beyond "chatbots" into true decision-making systems
- Traditional software patterns (like MVC, Observer) don't address the unique challenges of autonomous AI systems
- Real-world applications need agents that can handle complex, multi-step tasks
- Industry leaders (OpenAI, Anthropic, Google) are standardizing these patterns

What You'll Learn:

By the end of this document, you will understand:

- The philosophy behind agentic systems
- 5 fundamental design patterns for agents
- How to choose the right pattern for your problem
- How to implement agents in production
- How to debug and optimize agent behavior

SECTION 1: CONCEPTUAL FOUNDATIONS

1.1 What is an Agent? (Beginner Level)

Simple Definition

An **agent** is a software system that:

1. **Observes** the environment (input, context, data)
2. **Reasons** about what to do (decision-making)
3. **Acts** to achieve goals (takes actions, calls functions, makes API calls)
4. **Learns** from feedback (refines behavior)

Real-World Analogy: The Chef

Imagine a chef managing a kitchen:

- **Observes**: Available ingredients, orders, kitchen temperature
- **Reasons**: "I have 30 minutes, client wants pasta, I have fresh ingredients"
- **Acts**: Plans menu, cooks, plates dishes, serves
- **Learns**: "This sauce combo works well, I'll remember for next time"

An **agentic AI system** works similarly, but with code.

Key Difference: Agent vs. Traditional Software

Aspect	Traditional Software	Agentic AI
Decision-making	Hardcoded rules (if-then-else)	Learned from examples, contextual
Failure handling	Breaks or follows error flow	Reasons about failure, retries differently
Adaptability	Requires code changes	Adapts with new prompts/feedback
State	Simple variables	Complex context, reasoning history
Scalability	Add more servers	Improve prompts, better reasoning

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1.2 Why Do We Need Design Patterns? (Beginner Level)

The Problem

Without patterns, developers create agents that:

- Work for one specific task, but break on variations
- Cannot handle failures gracefully
- Are impossible to test
- Waste compute resources on unnecessary API calls
- Produce inconsistent results

The Solution: Design Patterns

Design patterns provide:

- **Reusability**: Solve similar problems with proven approaches
- **Testability**: Clear inputs, outputs, and failure modes
- **Scalability**: Works for 1 task, 100 tasks, or 1000 tasks
- **Maintenance**: Future developers understand the architecture
- **Performance**: Optimized for LLM token usage and latency

Pattern vs. Framework

- **Pattern** = Conceptual blueprint (like "use a loop with a check condition")
- **Framework** = Concrete implementation (like LangChain, LlamaIndex)

You need both: understand the pattern, then implement with a framework.

1.3 Historical Context: How We Got Here (Intermediate Level)

Evolution of AI Systems

Phase 1: Rule-Based Systems (1980s-2000s)

- Explicit rules: IF (symptom = fever) THEN (diagnosis = cold)
- Limitation: Couldn't generalize, required thousands of rules

Phase 2: Machine Learning (2000s-2010s)

- Learned patterns from data
- Limitation: Needed labeled data, couldn't reason about unfamiliar scenarios

Phase 3: Deep Learning / Neural Networks (2010s-2020s)

- Massive improvement in pattern recognition
- Limitation: "Black box" - couldn't explain decisions

Phase 4: Foundation Models & Transformers (2020-present)

- LLMs with in-context learning (few-shot prompting)
- Limitation: Single forward pass, no iteration or planning

Phase 5: Agentic Systems (2023-present) ← YOU ARE HERE

- LLMs + looping + external tools + state management
- Advantage: Can reason, plan, execute, and adapt
- New challenge: How do we architect this reliably?

1.4 Core Concepts You Must Understand (Intermediate Level)

1.4.1 The Observation-Reasoning-Action Loop

Every agent follows this cycle:

1.4.2 Agent Memory (Context Management)

Agents need memory to:

- Track what they've already tried
- Remember past decisions
- Build context for reasoning

Types of Memory:

1. **Short-term**: Current conversation, recent actions (in prompt)
2. **Long-term**: Learned patterns, user preferences (database)
3. **Tool history**: What functions were called, results received

1.4.3 Tools/Functions (The Agent's Capabilities)

An agent is only as powerful as its available tools:

```
Tools Available:
├── calculator()       → Can do math
├── web_search()      → Can browse internet
├── database_query()  → Can access data
├── email_send()      → Can communicate
└── file_read()       → Can access files

Agent's Capability = Quality of LLM + Diversity of Tools
```

Poor example: Giving a math agent a calculator but no access to data
Good example: Giving an analyst tools for web search, SQL, and visualization

1.4.4 Reasoning Frameworks

The LLM needs a system to think through problems:

Linear Reasoning (Simple)

```
Question → Think → Answer
```

Chain of Thought (Better)

```
Question → Step 1 → Step 2 → Step 3 → Answer
          (intermediate reasoning shown)
```

Tree of Thought (Best)

```
           Initial Problem
                 ↓
          ┌─ Branch A
          ├─ Branch B  (explore multiple paths)
          └─ Branch C
              ↓
           Best Solution
```

SECTION 2: CORE ARCHITECTURE & DEEP TECHNICAL EXPLANATION

2.1 The Agent's Brain: How LLMs Make Decisions (Intermediate Level)

The Token-Level Reality

When you ask an LLM to make a decision, here's what happens:

```
Input Tokens → Transformer Layers → Output Tokens (Decision)
               (50+ layers)

Example:
Input: "I need to book a flight. My dates are March 15-20."
       ↓ [Tokenized into ~20 tokens]
       ↓ [Processed through 80 transformer layers]
       ↓ [Attention mechanism: which tokens matter most?]
Output: I should use the flight_search() tool
        with parameters: date_start="2024-03-15",
                         date_end="2024-03-20"
```

The Prompt as a "Program"

Your prompt is essentially a program that tells the LLM:

```
System Prompt (Instructions) = Variable initialization
Few-shot Examples = Training data in real-time
User Query = Current input
Tool Definitions = Available functions
Context = State/memory
```

Example prompt architecture:

```
=== SYSTEM ===
You are a travel booking assistant. Be helpful and concise.

=== AVAILABLE TOOLS ===
1. flight_search(origin, destination, date_start, date_end) → list[Flight]
2. hotel_search(city, date_start, date_end) → list[Hotel]
3. book_flight(flight_id, passenger_details) → booking_confirmation

=== REASONING STYLE ===
Think step-by-step. First understand requirements, then take action.

=== USER CONTEXT ===
User preferences: prefers early morning flights, budget budget-conscious
Recent bookings: NYC to LA (Jan), Paris to Rome (Feb)

=== CURRENT QUERY ===
User: "Book me a flight from LA to NYC for March 15"

=== EXPECTED OUTPUT ===
Return your action as valid JSON:
{"action": "flight_search", "params": {"origin": "LAX", ...}}
```

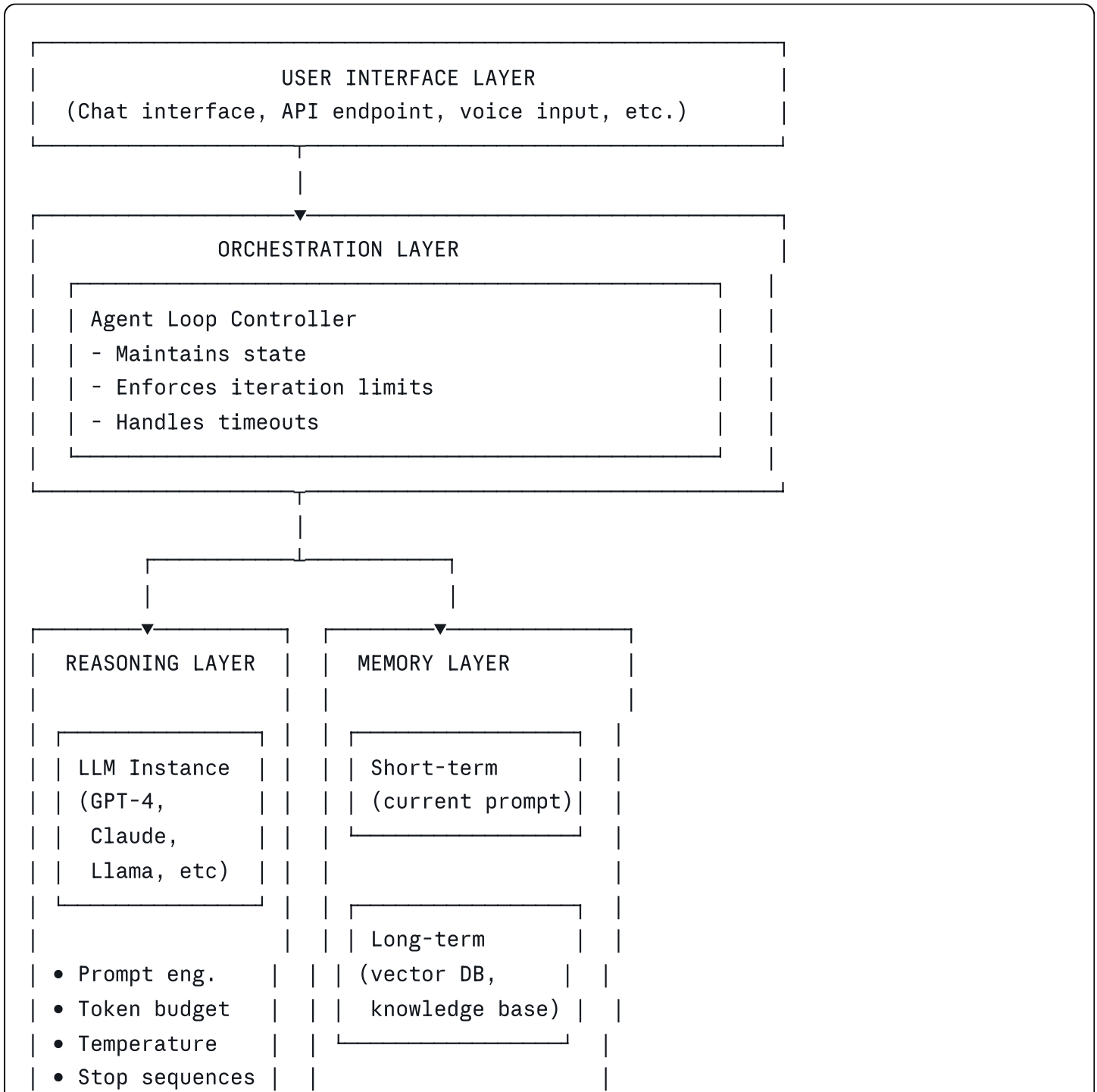
Attention Mechanism: Why Agents Work

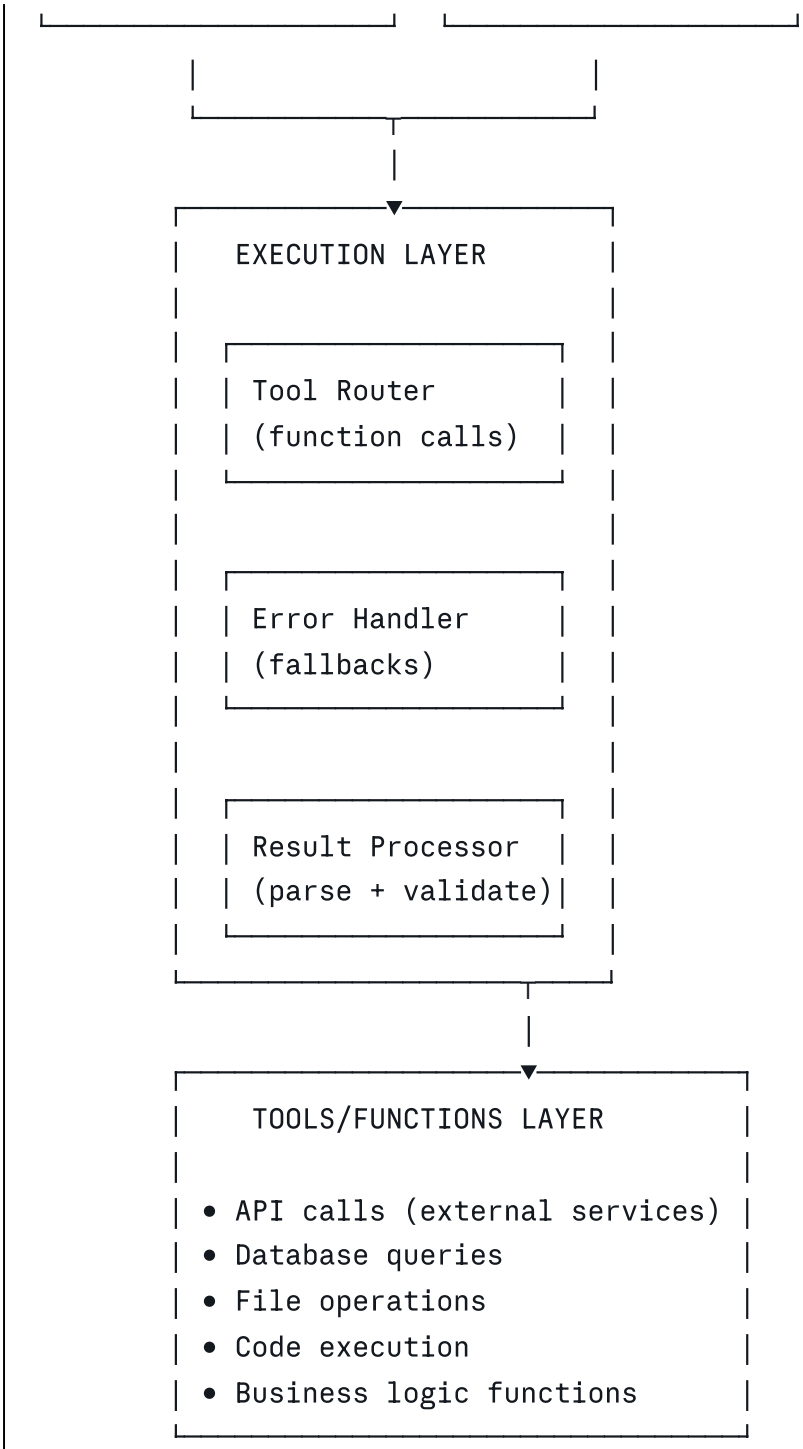
The transformer's attention mechanism allows the LLM to:

- Focus on relevant parts of the input
- Ignore irrelevant information
- Make connections between distant tokens

This is why agents work: **The LLM can attend to tool definitions, remember past actions, and reason about next steps.**

2.2 The Architecture Stack (Intermediate to Advanced)





2.3 The State Machine (Advanced)

Every agent operates as a finite state machine:

```
START
  |
  ├──→ READY (waiting for input)
  |
  └──→ PROCESSING
       |
       ├──→ OBSERVING (gathering information)
       |
       ├──→ REASONING (LLM thinking)
       |
       ├──→ DECIDING (which tool to use?)
       |    ├─ If need more info → EXECUTING
       |    ├─ If can answer → RESPONDING
       |    └─ If stuck → ERROR HANDLING
       |
       ├──→ EXECUTING (tool/function call)
       |
       ├──→ REFLECTING (evaluating result)
       |    ├─ If success → RESPONDING or LOOP
       |    └─ If failure → ERROR HANDLING
       |
       └──→ RESPONDING (output to user)
            |
            └──→ READY (loop back)

ERROR STATES:
├─ MAX_ITERATIONS (infinite loop prevention)
├─ TIMEOUT (too slow)
├─ TOOL_ERROR (function call failed)
├─ INVALID_FORMAT (LLM returned malformed JSON)
└─ CONTEXT_LENGTH (ran out of tokens)
```

2.4 Token Economy & Cost Optimization (Advanced)

The Token Budget Reality

```
Every LLM API call costs tokens:
- Input tokens:  $0.5 per 1M tokens (GPT-4)
- Output tokens: $15 per 1M tokens (GPT-4)

An agent loop might look like:
┌──────────────────────────────────────────────────┐
| Iteration 1:                                     |
| - Send: prompt + context (2000 tokens) → $0.001  |
| - Receive: decision (500 tokens) → $0.0075       |
| - Execute: tool call, get result (1000 tokens)   |
|                                                  |
| Iteration 2:                                     |
| - Send: old context + result (3500 tokens)       |
| - Receive: decision (300 tokens)                 |
|                                                  |
| Total: ~7300 tokens = $0.11                      |
|        (10 iterations could cost $1+)            |
└──────────────────────────────────────────────────┘
```

Cost Optimization Strategies

1. Prompt Compression

- Remove redundant context
- Use summaries instead of full history
- Prune irrelevant tool definitions

2. Intelligent Looping

- Set maximum iterations (prevent runaway agents)
- Use faster, cheaper models for intermediate steps
- Batch operations when possible

3. Caching

- Cache system prompts
- Store frequent results
- Use embeddings for semantic caching

4. Model Selection

```
For different agent types:
├─ Simple decisions → gpt-3.5-turbo (cheap, fast)
├─ Complex reasoning → gpt-4 (powerful, expensive)
├─ Local deployment → Llama 2, Mistral (free, slow)
└─ Edge cases → Mixture of experts (specialized)
```

SECTION 3: THE FIVE MAJOR AGENTIC PATTERNS

Pattern #1: THE LOOP PATTERN (Simplest)

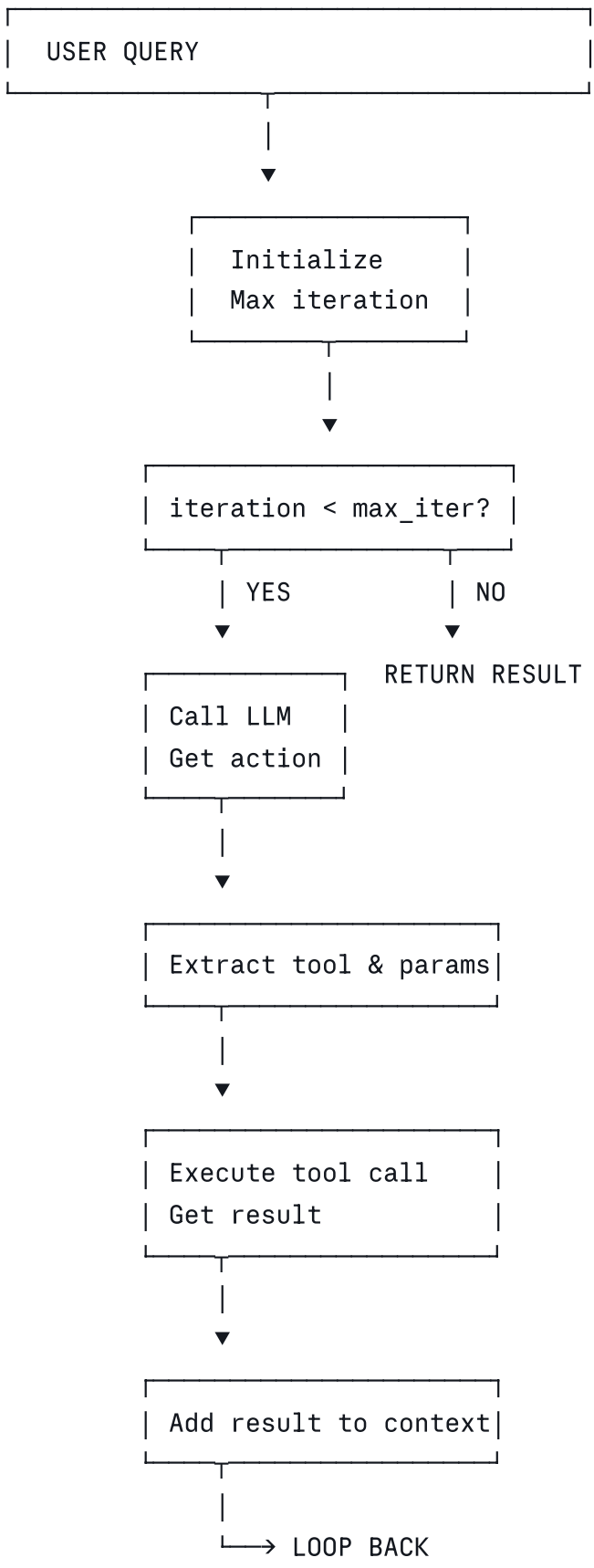
What is It?

The agent runs in a simple loop: observe → reason → act → check → repeat.

When to Use:

- Simple, single-task agents
- Clear success criteria
- Limited external dependencies

Architecture:



Code Example:

python

```python
def simple_loop_agent(query: str, max_iterations: int = 5):
    context = f"User query: {query}"

    for i in range(max_iterations):
        # OBSERVE: Get LLM decision
        response = llm.complete(
            prompt=f"{SYSTEM_PROMPT}\n{context}",
            tools=AVAILABLE_TOOLS
        )

        action = parse_action(response)

        # Check if done
        if action.type == "respond":
            return action.content

        # ACT: Execute tool
        try:
            result = execute_tool(action.tool_name, action.params)
        except Exception as e:
            result = f"Error: {str(e)}"

        # REFLECT: Add to context
        context += f"\nTool: {action.tool_name}\nResult: {result}"

    return "Max iterations reached"

# Usage:
answer = simple_loop_agent("What's the weather in NYC?")
```

Pros & Cons:

Pros	Cons
Simple to implement	Can loop forever
Easy to debug	Wastes tokens on long conversations
Transparent (clear flow)	No planning ahead
Good for single tasks	Fails on complex, multi-step problems

Pattern #2: THE PLANNING PATTERN (Think-Then-Do)

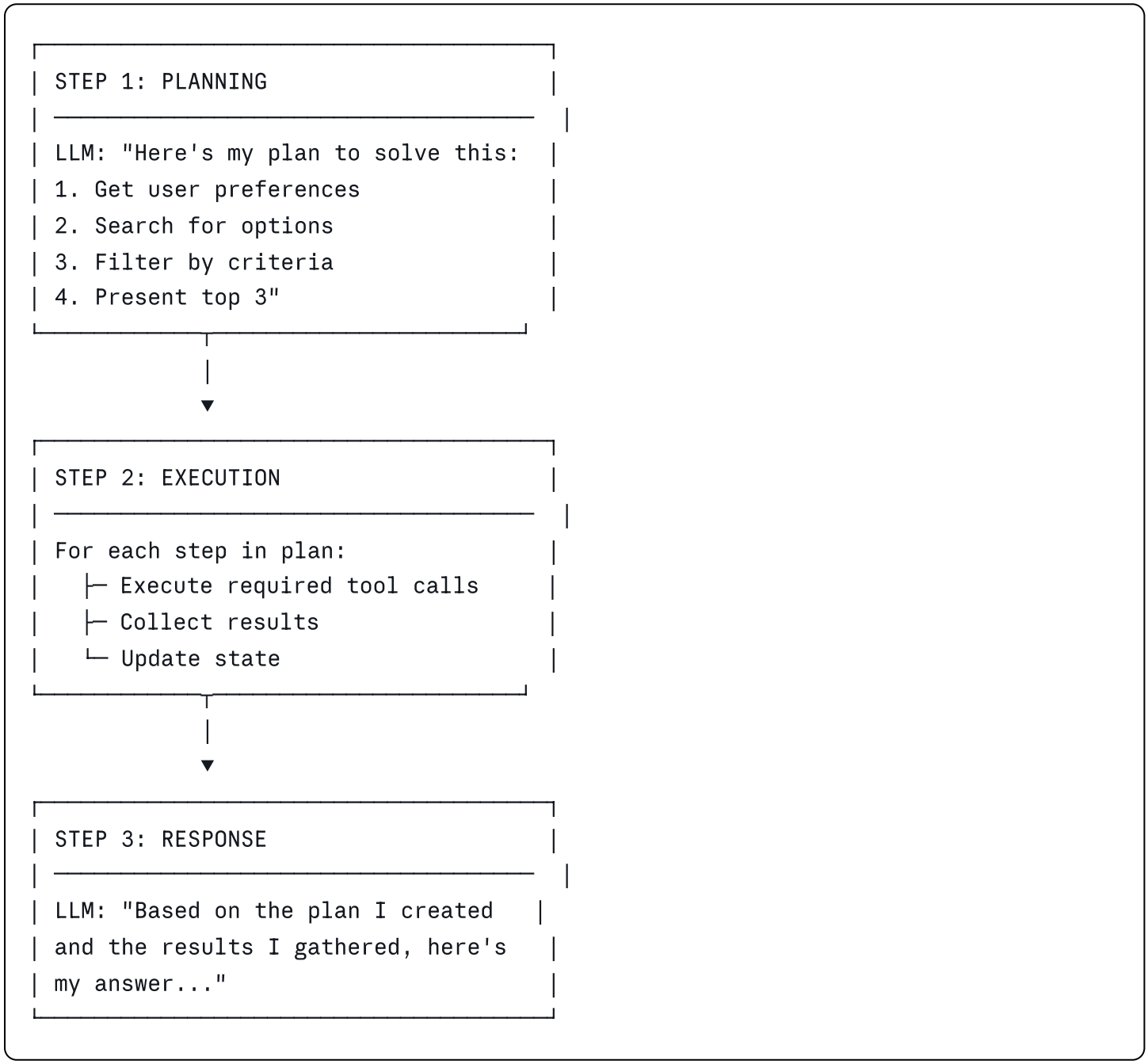
What is It?

Before acting, the agent creates a plan. Then it executes the plan step-by-step.

When to Use:

- Complex multi-step tasks
- Need to reduce token waste
- Want predictable behavior

Architecture:



Code Example:

```
python
```

```python
def planning_agent(query: str):
    # Step 1: CREATE PLAN
    plan_prompt = f"""
    User wants: {query}

    Create a detailed plan with numbered steps.
    Format:
    1. [Step description]
    2. [Step description]
    ...

    Plan:
    """

    plan_response = llm.complete(plan_prompt)
    steps = parse_plan(plan_response)  # Extract numbered steps

    # Step 2: EXECUTE PLAN
    results = {}
    for step_num, step in enumerate(steps):
        step_prompt = f"""
        Current task: {step}
        Previous results: {results}

        What tool should I use? Return JSON with tool_name and params.
        """

        action = llm.complete(step_prompt)
        action = parse_json(action)

        # Execute the tool
        result = execute_tool(action['tool_name'], action['params'])
        results[f"step_{step_num}"] = result

    # Step 3: RESPOND
    final_prompt = f"""
    Original request: {query}
    Plan executed: {steps}
    Results obtained: {results}

    Synthesize a final answer for the user.
    """

    return llm.complete(final_prompt)

# Usage:
```

```
answer = planning_agent(
    "Find me a hotel in Paris for March 15-20, 4-star, under $200/night"
)
```

Pros & Cons:

Pros	Cons
Reduces token waste	Plan might be wrong
Predictable behavior	Requires plan validation
Clear execution path	Less adaptive
Better for humans to review	Needs prompt engineering

Pattern #3: THE HIERARCHICAL PATTERN (Manager-Worker)

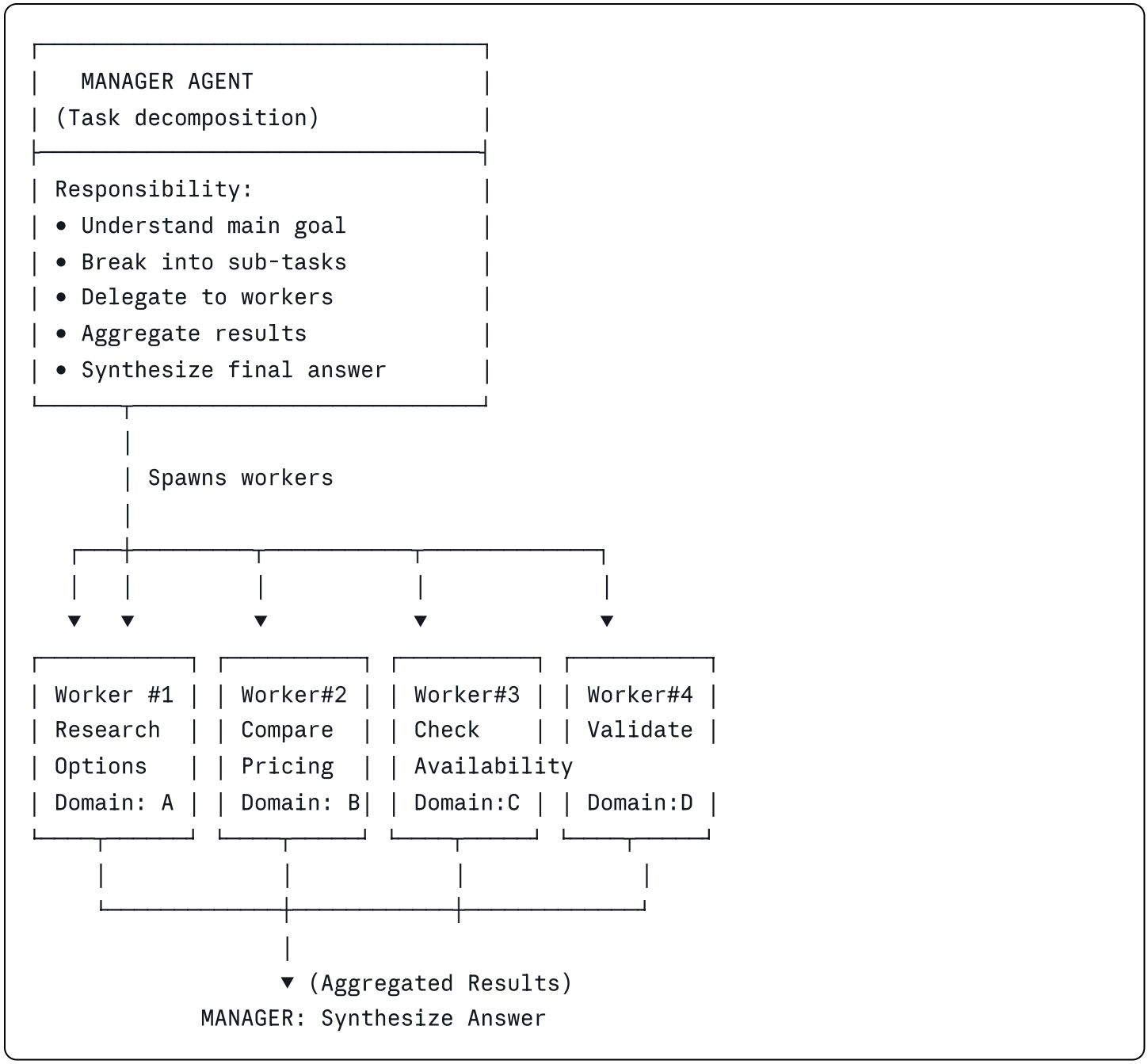
What is It?

A manager agent decomposes the task into sub-tasks and delegates to worker agents.

When to Use:

- Very complex problems
- Multiple independent sub-problems
- Specialized agents for different domains
- Scalability needed

Architecture:



Code Example:

python

```python
class WorkerAgent:
    def __init__(self, domain: str, tools: list):
        self.domain = domain
        self.tools = tools

    def execute(self, task: str) -> str:
        prompt = f"""
        You are a {self.domain} expert.
        Task: {task}
        Use your specialized knowledge to complete this.
        """
        return llm.complete(prompt, tools=self.tools)

class ManagerAgent:
    def __init__(self):
        self.workers = {
            "research": WorkerAgent("research", RESEARCH_TOOLS),
            "pricing": WorkerAgent("pricing", PRICING_TOOLS),
            "availability": WorkerAgent("availability", AVAILABILITY_TOOLS),
            "validation": WorkerAgent("validation", VALIDATION_TOOLS),
        }

    def solve(self, query: str):
        # Step 1: Decompose
        decompose_prompt = f"""
        User query: {query}

        Break this into independent sub-tasks.
        Return JSON:
        {{
            "subtasks": [
                {{"worker": "research", "task": "..."}},
                {{"worker": "pricing", "task": "..."}},
                ...
            ]
        }}
        """

        decomposition = llm.complete(decompose_prompt)
        subtasks = parse_json(decomposition)["subtasks"]

        # Step 2: Delegate
        results = {}
        for subtask in subtasks:
            worker_type = subtask["worker"]
            worker_task = subtask["task"]
```

```
            result = self.workers[worker_type].execute(worker_task)
            results[worker_type] = result

        # Step 3: Synthesize
        synthesis_prompt = f"""
        Original query: {query}

        Results from specialized agents:
        {json.dumps(results, indent=2)}

        Synthesize these into a comprehensive answer.
        """

        return llm.complete(synthesis_prompt)

# Usage:
manager = ManagerAgent()
answer = manager.solve("Find best investment opportunity in tech sector")
```

Pros & Cons:

Pros	Cons
Handles very complex problems	Higher latency (parallel calls)
Modular and scalable	More expensive (multiple LLM calls)
Easy to add new specialists	Hard to debug
Natural decomposition	Requires careful task design

Pattern #4: THE REFLECTION PATTERN (Self-Improvement Loop)

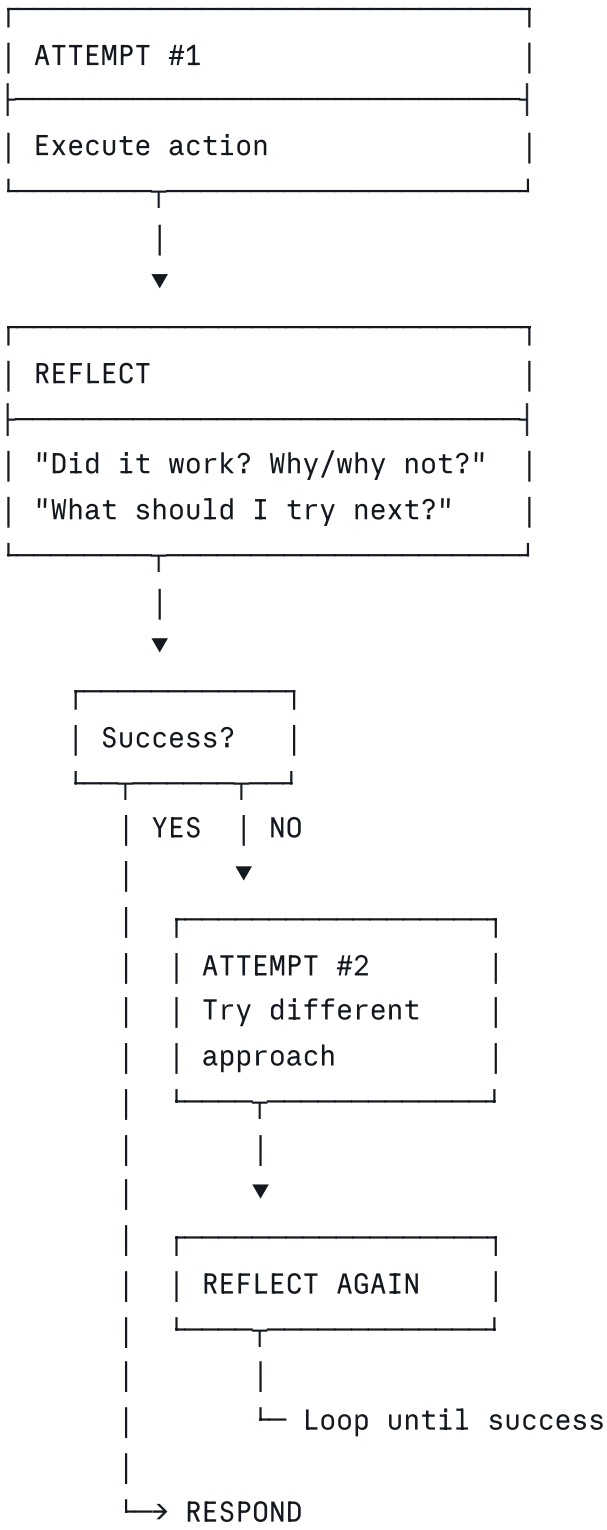
What is It?

After taking action, the agent reflects on whether it worked. If not, it tries a different approach.

When to Use:

- When success is uncertain
- Need iterative refinement
- Want agents to "learn" within a session
- Quality critical applications

Architecture:



Code Example:

```python
```

```python
def reflection_agent(query: str, max_attempts: int = 3):
    context = f"User query: {query}"

    for attempt in range(max_attempts):
        # ACT: Get LLM suggestion
        action_prompt = f"""
        {context}

        Previous attempts: {attempt}

        What's your next action? Return JSON.
        """

        action = parse_json(llm.complete(action_prompt))

        # EXECUTE
        try:
            result = execute_tool(action['tool'], action['params'])
            execution_success = True
        except Exception as e:
            result = str(e)
            execution_success = False

        # REFLECT
        reflect_prompt = f"""
        Attempted action: {action}
        Result: {result}

        Did this successfully address the user's query?
        Return JSON:
        {{
            "success": true/false,
            "reasoning": "...",
            "next_step": "..."
        }}
        """

        reflection = parse_json(llm.complete(reflect_prompt))

        if reflection['success']:
            return result  # Success!

        # Update context for next attempt
        context += f"\nAttempt {attempt + 1}: {action['tool']}\nResult: {result}\nIs

    return "Unable to complete after max attempts"
```

```
# Usage:
answer = reflection_agent("Translate this poem to Spanish while preserving meter")
```

Pros & Cons:

Pros	Cons
Self-correcting	Uses more tokens (multiple attempts)
Better quality	Takes longer
Handles ambiguity well	Can get stuck in loops
More robust	Needs good reflection prompts

Pattern #5: THE MULTI-AGENT PATTERN (Collaborative Teams)

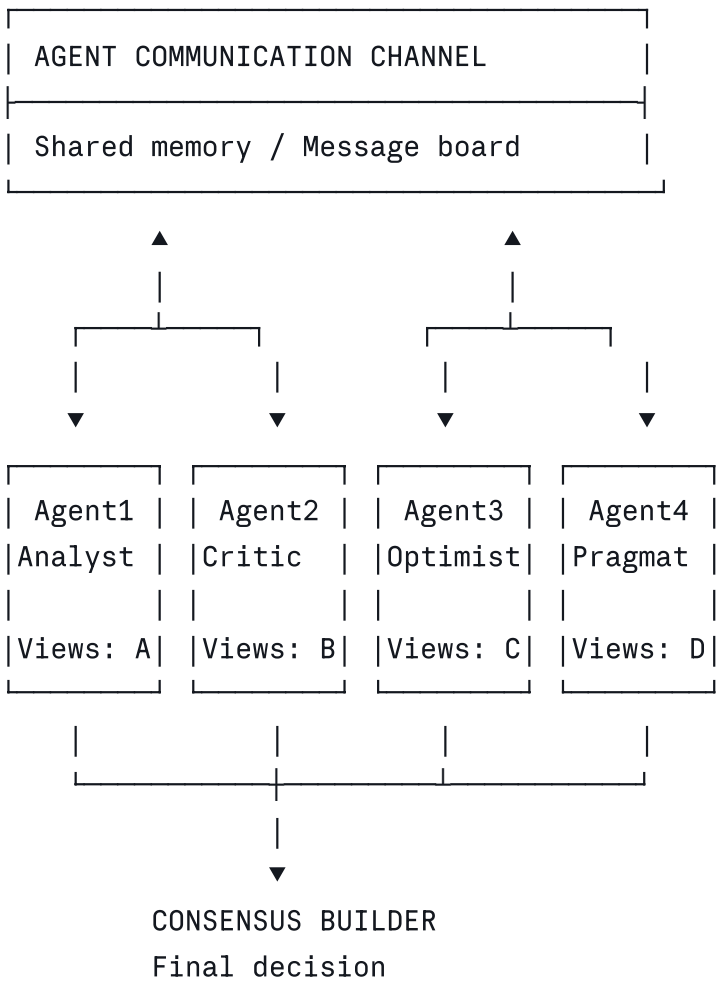
What is It?

Multiple agents work together, communicating and coordinating to solve a problem.

When to Use:

- Problems requiring different perspectives
- Need healthy debate/discussion
- Complex stakeholder scenarios
- Want emergent behavior

Architecture:



Code Example:

python

```
class TeamAgent:
    def __init__(self, name: str, role: str, perspective: str):
        self.name = name
        self.role = role
        self.perspective = perspective

    def contribute(self, discussion_context: str) -> str:
        prompt = f"""
        You are {self.name}, a {self.role}.
        Your perspective: {self.perspective}

        Previous discussion:
        {discussion_context}

        What's your perspective? Be concise and specific.
        """
        return llm.complete(prompt)

class MultiAgentTeam:
    def __init__(self):
        self.agents = [
            TeamAgent("Alice", "Data Analyst",
                      "Focus on facts and data"),
            TeamAgent("Bob", "Devil's Advocate",
                      "Question assumptions"),
            TeamAgent("Carol", "Business Strategist",
                      "Think about ROI and strategy"),
            TeamAgent("David", "Risk Manager",
                      "Identify potential problems"),
        ]

    def deliberate(self, topic: str, rounds: int = 3):
        discussion = f"Topic: {topic}\n\n"

        for round_num in range(rounds):
            discussion += f"=== ROUND {round_num + 1} ===\n"

            for agent in self.agents:
                contribution = agent.contribute(discussion)
                discussion += f"{agent.name}: {contribution}\n"

        # Reach consensus
        consensus_prompt = f"""
        Discussion on {topic}:
        {discussion}
```

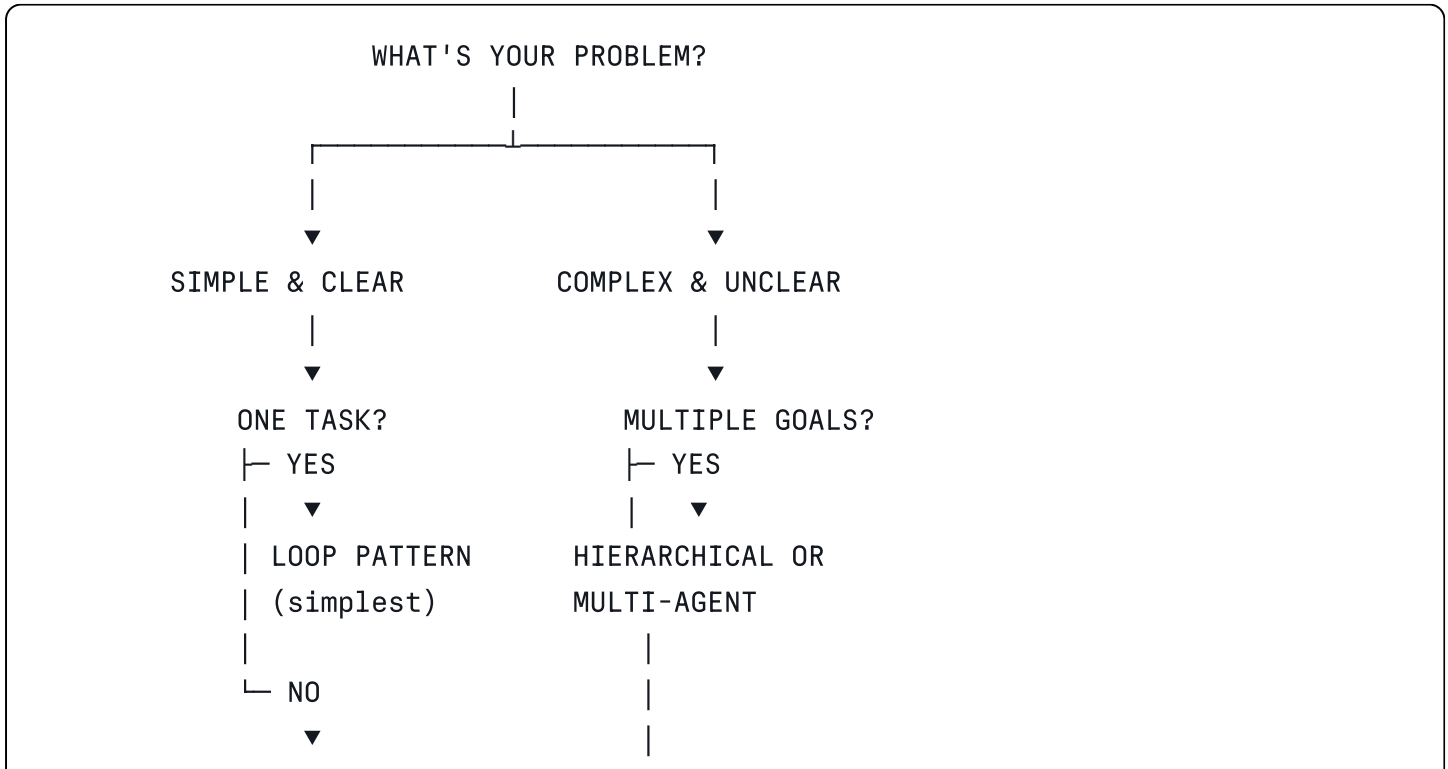
```
        Summarize the key points and reach a balanced conclusion.
        """

        return llm.complete(consensus_prompt)

# Usage:
team = MultiAgentTeam()
decision = team.deliberate(
    "Should we launch feature X?",
    rounds=2
)
```

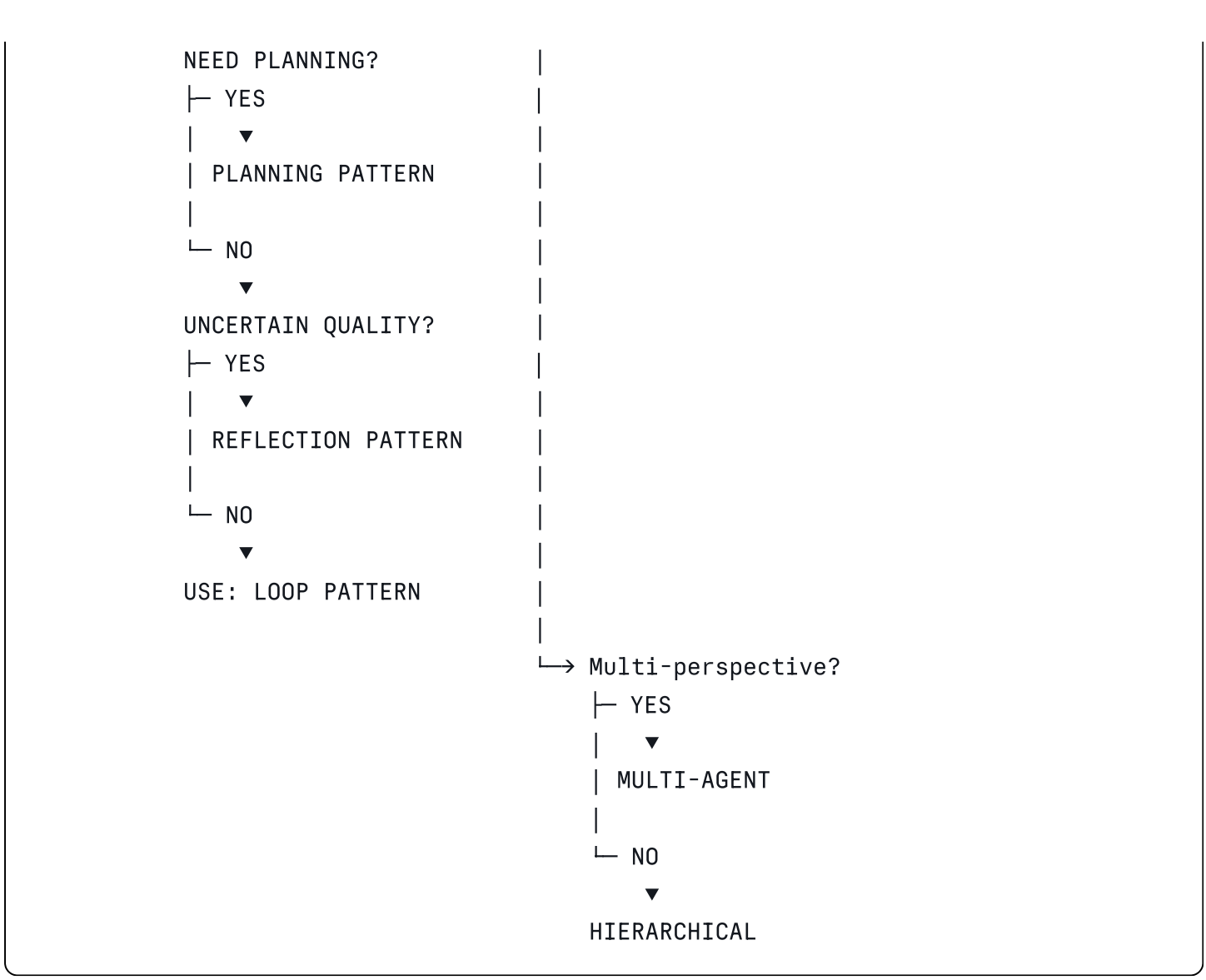
Pros & Cons:

Pros	Cons
Captures multiple viewpoints	Hardest to implement
Better decisions through debate	Expensive (many LLM calls)
Emergent insights	Hard to coordinate
More robust analysis	Can have conflicting conclusions

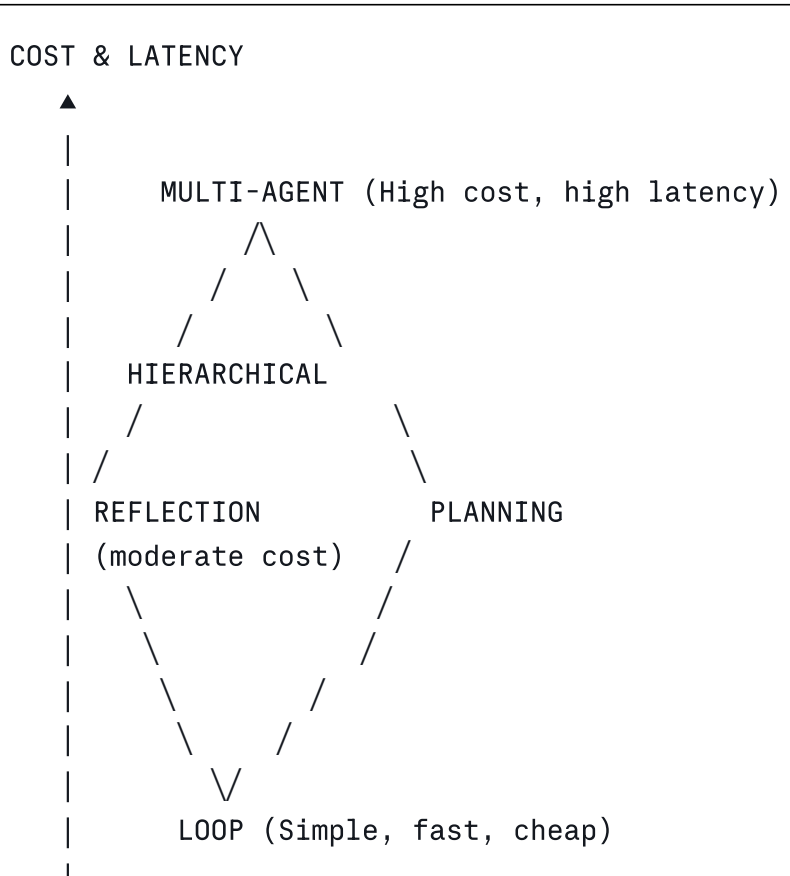
SECTION 4: VISUAL FRAMEWORKS & DIAGRAMS

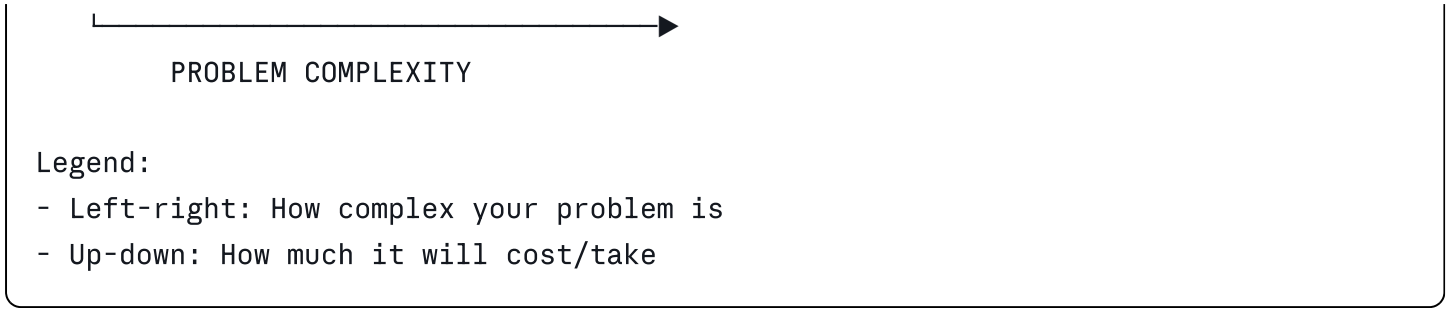
4.1 Decision Tree: Which Pattern to Use?



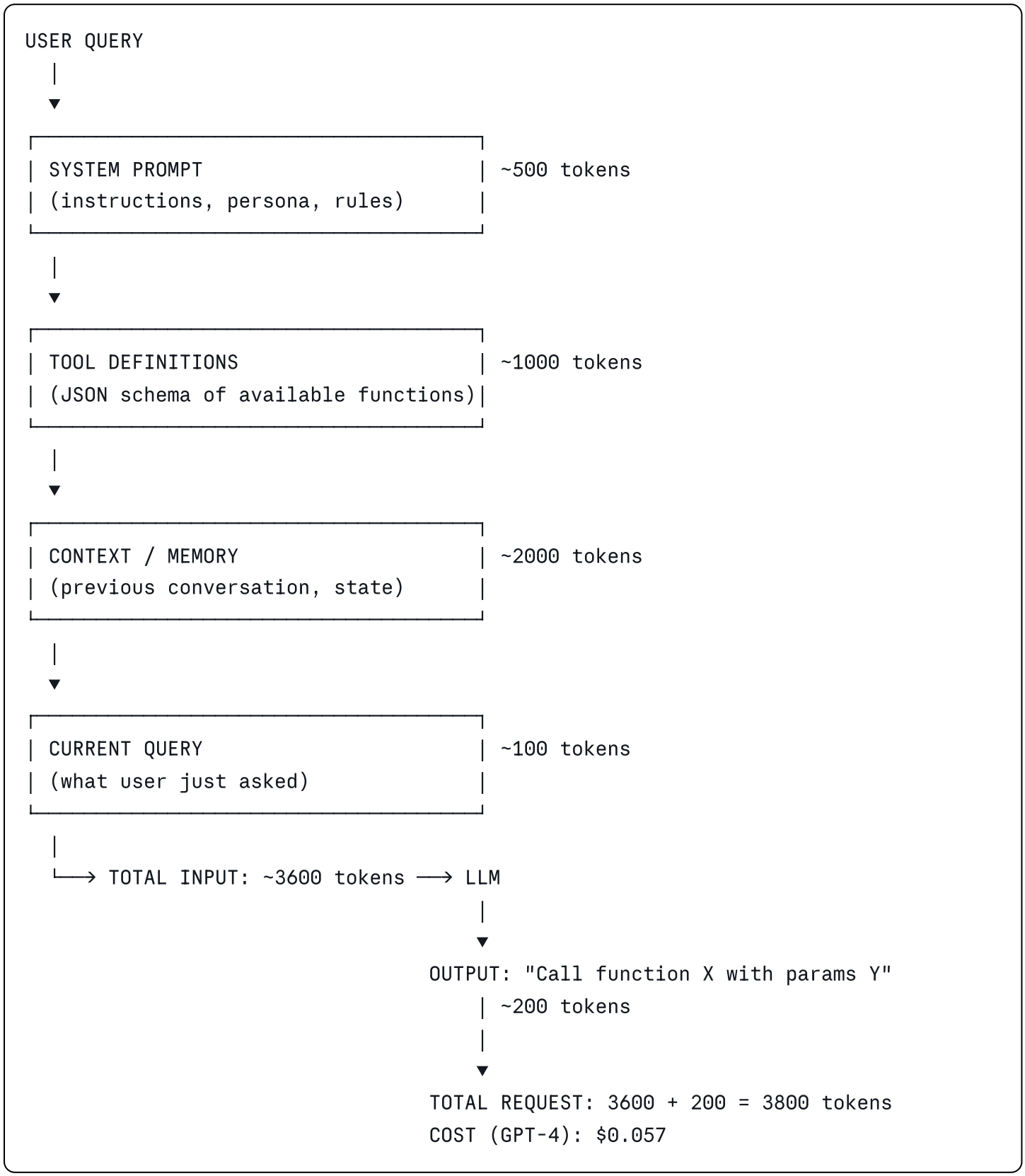


4.2 Agent Complexity vs. Cost Trade-off

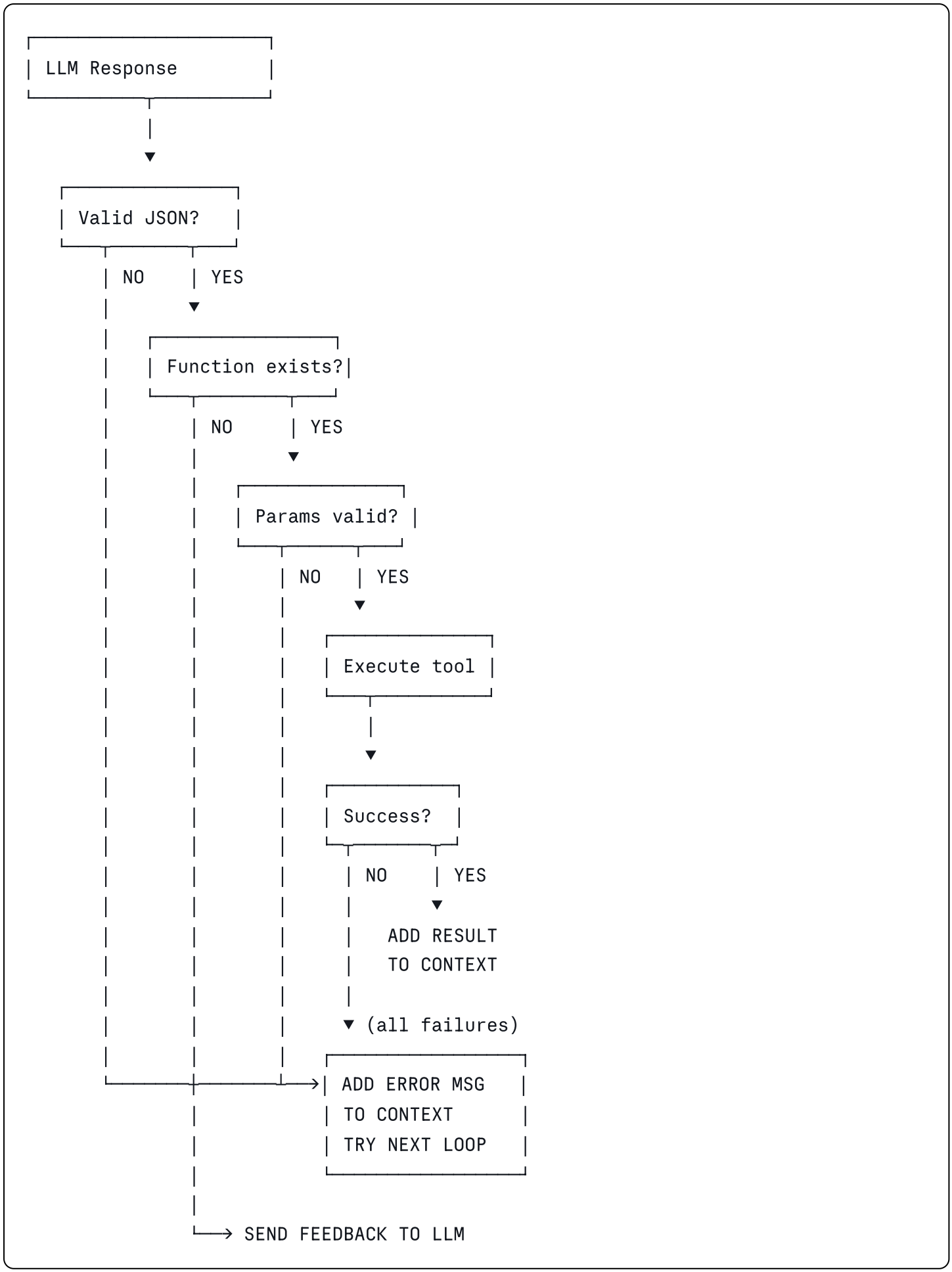




4.3 Token Flow Diagram: Understanding What's Being Sent



4.4 Error Recovery Flow



SECTION 5: PRACTICAL CODE EXAMPLES

5.1 Complete Working Example: Customer Service Agent

python

```python
from typing import List, Dict, Any
import json
from dataclasses import dataclass
from enum import Enum

# ============================================================================
# DATA STRUCTURES
# ============================================================================

class ToolType(Enum):
    RESPOND = "respond"
    SEARCH_KNOWLEDGE_BASE = "search_knowledge_base"
    GET_ORDER_STATUS = "get_order_status"
    INITIATE_REFUND = "initiate_refund"
    ESCALATE_TO_HUMAN = "escalate_to_human"

@dataclass
class ToolCall:
    tool: ToolType
    params: Dict[str, Any]

@dataclass
class Message:
    role: str  # "user" or "assistant"
    content: str

# ============================================================================
# MOCK LLM & TOOLS (In real world, these call actual APIs)
# ============================================================================

class MockLLM:
    """Simulates GPT/Claude behavior"""

    def __init__(self):
        self.conversation_history = []

    def complete(self, prompt: str, tools: List[Dict]) -> str:
        """
        In real implementation, this would call:
        - OpenAI API: client.chat.completions.create()
        - Anthropic API: client.messages.create()
        - Local LLM: llama.generate()

        For demo, we return hardcoded responses based on keywords.
        """
```

```
        # Simple keyword matching for demo
        if "refund" in prompt.lower():
            return json.dumps({
                "thinking": "Customer wants a refund. I should check order first.",
                "tool": "get_order_status",
                "params": {"order_id": "ORD-12345"}
            })
        elif "track" in prompt.lower() or "status" in prompt.lower():
            return json.dumps({
                "thinking": "Customer wants to know order status.",
                "tool": "search_knowledge_base",
                "params": {"query": "How do I track my order?"}
            })
        else:
            return json.dumps({
                "thinking": "I have enough info to respond.",
                "tool": "respond",
                "params": {"message": "I'm here to help!"}
            })

class Tools:
    """Execute various business functions"""

    @staticmethod
    def search_knowledge_base(query: str) -> str:
        """Search company's FAQ/knowledge base"""
        kb = {
            "shipping": "Orders typically arrive in 5-7 business days.",
            "refund": "We offer 30-day refunds for most items.",
            "tracking": "Use the tracking number sent to your email.",
        }

        for keyword, answer in kb.items():
            if keyword in query.lower():
                return answer

        return "I couldn't find information about that. Let me escalate."

    @staticmethod
    def get_order_status(order_id: str) -> str:
        """Check order status from database"""
        # Mock data
        orders = {
            "ORD-12345": "Shipped on March 10, arrives March 15",
            "ORD-67890": "Processing",
        }
```

```
        return orders.get(order_id, f"Order {order_id} not found")

    @staticmethod
    def initiate_refund(order_id: str, reason: str) -> str:
        """Process refund request"""
        return f"Refund initiated for {order_id}. Reason: {reason}. You'll see credi

    @staticmethod
    def escalate_to_human(issue: str) -> str:
        """Escalate to human agent"""
        return f"I'm escalating your issue to our human support team: '{issue}'. The

# ============================================================================
# AGENTIC SYSTEM - THE LOOP PATTERN
# ============================================================================

class CustomerServiceAgent:
    """
    An agent that handles customer service queries.
    Uses the LOOP PATTERN.
    """

    def __init__(self, max_iterations: int = 5, debug: bool = True):
        self.llm = MockLLM()
        self.tools_executor = Tools()
        self.max_iterations = max_iterations
        self.debug = debug
        self.conversation_history = []

    def _parse_tool_call(self, response_text: str) -> ToolCall:
        """Extract tool name and parameters from LLM response"""
        try:
            data = json.loads(response_text)
            tool_name = data.get("tool", "respond").upper()
            params = data.get("params", {})

            return ToolCall(
                tool=ToolType[tool_name],
                params=params
            )
        except json.JSONDecodeError:
            # Fallback if LLM doesn't return valid JSON
            return ToolCall(tool=ToolType.RESPOND, params={"message": response_text}

    def _execute_tool(self, tool_call: ToolCall) -> str:
        """Execute the tool and return result"""
```

```
        if tool_call.tool == ToolType.RESPOND:
            return tool_call.params.get("message", "How can I help?")

        elif tool_call.tool == ToolType.SEARCH_KNOWLEDGE_BASE:
            return self.tools_executor.search_knowledge_base(
                tool_call.params.get("query", "")
            )

        elif tool_call.tool == ToolType.GET_ORDER_STATUS:
            return self.tools_executor.get_order_status(
                tool_call.params.get("order_id", "")
            )

        elif tool_call.tool == ToolType.INITIATE_REFUND:
            return self.tools_executor.initiate_refund(
                order_id=tool_call.params.get("order_id", ""),
                reason=tool_call.params.get("reason", "")
            )

        elif tool_call.tool == ToolType.ESCALATE_TO_HUMAN:
            return self.tools_executor.escalate_to_human(
                tool_call.params.get("issue", "")
            )

        else:
            return "Unknown tool"

    def process(self, user_query: str) -> str:
        """
        Main agent loop.

        THE LOOP PATTERN:
        1. Observe (read user input)
        2. Reason (LLM decides action)
        3. Act (execute tool)
        4. Reflect (check if done)
        5. Loop or return answer
        """

        # Add user message to history
        self.conversation_history.append(
            Message(role="user", content=user_query)
        )

        if self.debug:
            print(f"\n{'='*60}")
            print(f"USER: {user_query}")
```

```
        print(f"{'='*60}")

    # ===== LOOP PATTERN =====
    for iteration in range(self.max_iterations):
        # STEP 1: OBSERVE - Build context
        context = self._build_context()

        # STEP 2: REASON - Get LLM decision
        llm_response = self.llm.complete(context, tools=[])

        if self.debug:
            print(f"\n[ITERATION {iteration + 1}]")
            print(f"LLM Response: {llm_response}")

        # Parse the response
        tool_call = self._parse_tool_call(llm_response)

        # STEP 3: ACT - Execute the tool
        if self.debug:
            print(f"Tool: {tool_call.tool.value}")
            print(f"Params: {tool_call.params}")

        result = self._execute_tool(tool_call)

        if self.debug:
            print(f"Result: {result}")

        # STEP 4: REFLECT - Check if we're done
        if tool_call.tool == ToolType.RESPOND:
            # Agent decided to respond, we're done
            return result

        # Add result to history for next iteration
        self.conversation_history.append(
            Message(role="assistant", content=f"Used {tool_call.tool.value}: {re
        )

    # Max iterations reached
    return "I've reached my thinking limit. Let me escalate this to a human agen

def _build_context(self) -> str:
    """Build the prompt context from conversation history"""
    context = "SYSTEM: You are a helpful customer service agent.\n\n"

    context += "AVAILABLE TOOLS:\n"
    context += "- search_knowledge_base(query): Search FAQ\n"
    context += "- get_order_status(order_id): Check order\n"
```

```
        context += "- initiate_refund(order_id, reason): Process refund\n"
        context += "- escalate_to_human(issue): Escalate\n"
        context += "- respond(message): Answer customer\n\n"

        context += "CONVERSATION HISTORY:\n"
        for msg in self.conversation_history:
            context += f"{msg.role.upper()}: {msg.content}\n"

        return context

# ============================================================================
# USAGE
# ============================================================================

def main():
    agent = CustomerServiceAgent(debug=True)

    # Test cases
    test_queries = [
        "I'd like to track my order",
        "Can I get a refund? I ordered item #12345 but changed my mind",
    ]

    for query in test_queries:
        response = agent.process(query)
        print(f"\n{'='*60}")
        print(f"AGENT FINAL RESPONSE: {response}")
        print(f"{'='*60}\n")

if __name__ == "__main__":
    main()
```

5.2 Planning Pattern Example: Recruitment Agent

python

```python
from typing import List
from dataclasses import dataclass

@dataclass
class JobRequirement:
    title: str
    required_skills: List[str]
    budget: int
    location: str

class RecruitmentPlanningAgent:
    """
    Uses PLANNING PATTERN:
    1. Analyze job requirements
    2. Create a recruitment plan
    3. Execute each step
    4. Synthesize results
    """

    def __init__(self):
        self.llm = MockLLM()

    def find_candidates(self, job: JobRequirement) -> str:
        # STEP 1: CREATE PLAN
        plan_prompt = f"""
        Job to fill: {job.title}
        Required skills: {', '.join(job.required_skills)}
        Budget: ${job.budget}
        Location: {job.location}

        Create a step-by-step recruitment plan. Return as numbered list:
        1. Search strategy
        2. Screening criteria
        3. Interview questions
        4. Final decision process
        """

        plan = self.llm.complete(plan_prompt, tools=[])
        print("=== RECRUITMENT PLAN ===")
        print(plan)

        # STEP 2: EXECUTE PLAN
        results = {}

        # Step 1: Search
        results['search'] = self._execute_search(job)
```

```
        # Step 2: Screen
        results['screening'] = self._execute_screening(results['search'], job)

        # Step 3: Interview
        results['interviews'] = self._execute_interviews(results['screening'])

        # Step 4: Decide
        results['decision'] = self._execute_decision(results['interviews'])

        # STEP 3: SYNTHESIZE
        synthesis_prompt = f"""
        Job: {job.title}

        Recruitment Results:
        - Initial candidates: {len(results['search'])}
        - Passed screening: {len(results['screening'])}
        - Interviewed: {len(results['interviews'])}
        - Recommended: {results['decision']}

        Provide final hiring recommendation.
        """

        final_rec = self.llm.complete(synthesis_prompt, tools=[])
        return final_rec

    def _execute_search(self, job) -> List[str]:
        """Step 1: Find candidates"""
        return ["Alice", "Bob", "Carol"]  # Mock

    def _execute_screening(self, candidates, job) -> List[str]:
        """Step 2: Screen candidates"""
        return candidates[:2]  # Mock

    def _execute_interviews(self, candidates) -> List[str]:
        """Step 3: Interview"""
        return candidates[:1]  # Mock

    def _execute_decision(self, interviewed) -> str:
        """Step 4: Make decision"""
        return interviewed[0] if interviewed else "No suitable candidate"
```

SECTION 6: HANDS-ON EXERCISES

Exercise 1: Build a Simple Loop Agent (Beginner)

Objective: Implement the simplest agentic pattern

Task: Create an agent that:

- Asks a user for their favorite programming language
- Searches a knowledge base for facts about that language
- Returns interesting facts

Code Template:

```python
def language_info_agent(user_input: str) -> str:
    """TODO: Implement the loop pattern"""

    context = f"User input: {user_input}"

    for i in range(3):  # Max 3 iterations
        # 1. OBSERVE: Build prompt
        prompt = f"Given this: {context}, what should I do?"

        # 2. REASON: Get LLM decision
        decision = call_llm(prompt)

        # 3. ACT: Execute
        # TODO: Implement tool execution

        # 4. REFLECT: Check if done
        # TODO: Check if success

        context += f"\nStep {i}: {decision}"

    return "Final answer"
```

Acceptance Criteria:

- [ ] Agent loops at least once
- [ ] Can call at least one tool
- [ ] Returns a sensible answer
- [ ] Handles edge cases (unknown language, etc.)

Exercise 2: Implement Error Handling (Intermediate)

Objective: Make agents more robust

Task: Modify Exercise 1 to:

- Gracefully handle invalid tool responses
- Retry failed tool calls
- Provide fallback answers

Key Points:

```python
def robust_agent(query: str) -> str:
    max_retries = 3
    retry_count = 0

    while retry_count < max_retries:
        try:
            # Try to execute tool
            result = execute_tool(...)
            return result
        except ToolError as e:
            # Handle specific tool errors
            retry_count += 1
            # TODO: Log error, try different approach
        except Exception as e:
            # Handle unexpected errors
            return "I encountered an error. Please try again."
```

Exercise 3: Multi-Step Agent (Intermediate)

Objective: Build a real-world multi-step workflow

Task: Create a weather recommendation agent that:

1. Gets current weather for a location
2. Checks if it matches user preferences
3. Recommends activities
4. Provides safety warnings

Workflow:

```
User: "What should I do in NYC tomorrow?"
  ↓
Agent: "Let me get the weather forecast"
  ↓
Agent: "Tomorrow is sunny, 72°F"
  ↓
Agent: "Since you like outdoor activities, I recommend..."
  ↓
Agent: "No safety warnings for those activities"
  ↓
Return: Recommendation
```

Exercise 4: Reflection Agent (Advanced)

Objective: Implement self-correction

Task: Create an agent that:

- Attempts to solve a math problem
- Reflects on whether the answer is reasonable
- Corrects itself if needed

Hints:

```python
def self_correcting_agent(problem: str) -> str:
    for attempt in range(3):
        answer = llm.solve(problem)

        is_reasonable = llm.verify(f"Is {answer} reasonable for {problem}?")

        if is_reasonable:
            return answer
        else:
            problem += "\nPrevious wrong answer: " + answer

    return "Unable to solve"
```

Exercise 5: Mini-Project - Trip Planner Agent (Advanced)

Objective: Combine all patterns

Requirements:

- Takes user input: "Plan a 3-day trip to Japan for 2 people with $3000 budget"
- Breaks down into: flights, hotels, activities
- Uses multiple tools: flight API, hotel API, activity API
- Validates budget
- Returns structured itinerary

Bonus:

- Add reflection: "Does this fit the budget?"
- Add planning: "What's the best order of activities?"
- Add error handling: Handle API failures gracefully

SECTION 7: COMMON MISTAKES & MISCONCEPTIONS

Mistake #1: "Agents Will Solve Anything If I Use Better Prompts"

❌ **Wrong Thinking:** "If the agent fails, I just need to write a better prompt."

✅ **Reality:**

- Prompts help, but can't compensate for bad architecture
- Problem is often not the prompt, but:
  - Wrong tools available
  - Infinite loops
  - Token waste
  - Poor error handling

Example:

```
WRONG:
Prompt: "Please be helpful and solve this complex problem well!"
Result: Agent goes into infinite loop

RIGHT:
- Add max_iterations limit
- Provide specific tools
- Add reflection step
```

Mistake #2: "Agents Need Longer Context to Work Better"

❌ **Wrong Thinking**: "Longer prompts = better reasoning"

✅ **Reality**:

- LLMs have a context window limit (4K, 8K, 128K tokens)
- More context = more cost and slower responses
- Quality of context matters more than quantity

Example:

```
WRONG: Include 100K tokens of conversation history
Result: $15 per query, slow response, poor reasoning

RIGHT: Summarize, compress, keep only relevant context
Result: $0.50 per query, fast response, better reasoning
```

Mistake #3: "My Agent Works Locally, So It's Production-Ready"

❌ **Wrong Thinking**: "If it works on my laptop, it can handle production traffic"

✅ **Reality**: You need to handle:

- **Concurrent requests**: 1 user vs. 1000 users
- **Rate limits**: API throttling
- **Failures**: Network timeouts, API errors
- **Monitoring**: Know when things break
- **Cost**: Local testing is cheap; production is expensive

Mistake #4: "Agentic = LLM Calling Tools"

❌ **Wrong Thinking**: "An agent is just an LLM with function calling"

✅ **Reality**: Agentic is a full system:

- Reasoning (LLM)
- Memory (context management)
- Execution (tool calling)
- Reflection (did it work?)
- Error handling (what if it fails?)
- Optimization (cost, speed)

Mistake #5: "I Should Use Multi-Agent for Everything"

❌ **Wrong Thinking**: "More agents = more intelligence"

✅ **Reality**:

- Multi-agent is powerful but expensive
- Use simplest pattern that works
- Start with loop → planning → hierarchical → multi-agent

Pattern Selection:

```
Simple task?          → Loop Pattern
Multi-step task?      → Planning Pattern
Very complex?         → Hierarchical Pattern
Need debate?          → Multi-Agent Pattern
Uncertain quality?    → Reflection Pattern
```

Mistake #6: "LLM Hallucinations Don't Matter for My Agent"

❌ **Wrong Thinking**: "Agents will fix hallucinations automatically"

✅ **Reality**: LLMs still hallucinate:

- Invent function names
- Imagine data that doesn't exist
- Forget facts from context

Solutions:

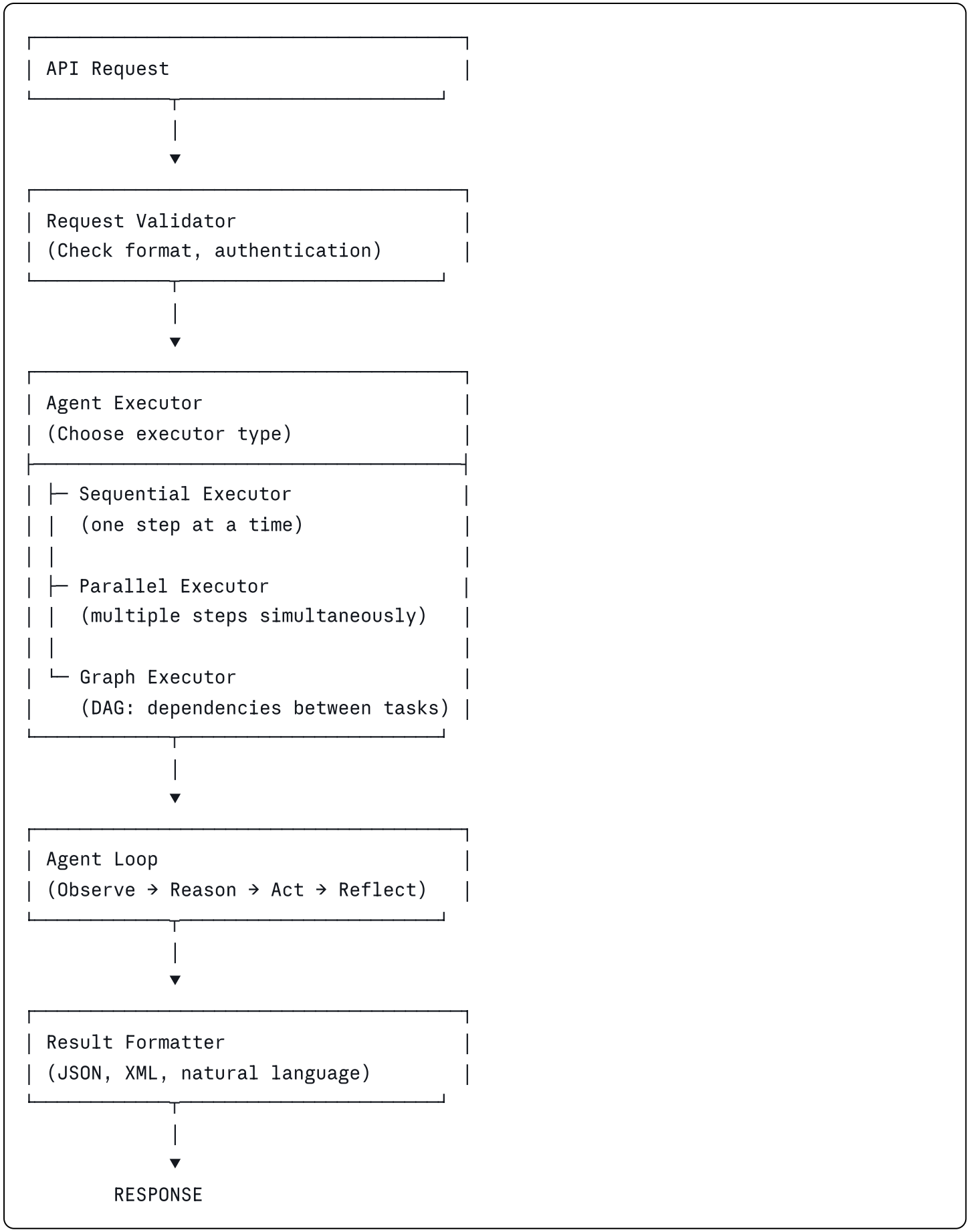
```python
```

```
# Validate tool calls
available_tools = ["tool_a", "tool_b"]
if llm_response['tool'] not in available_tools:
    # Error, don't execute
    ask_llm_to_try_again()

# Verify results
result = execute_tool()
if not is_reasonable(result):
    # Error, try again
    llm.reflect("That result seems wrong, why?")
```

SECTION 8: ADVANCED CONCEPTS

8.1 Agent Architectures in Production

The Executor Pattern



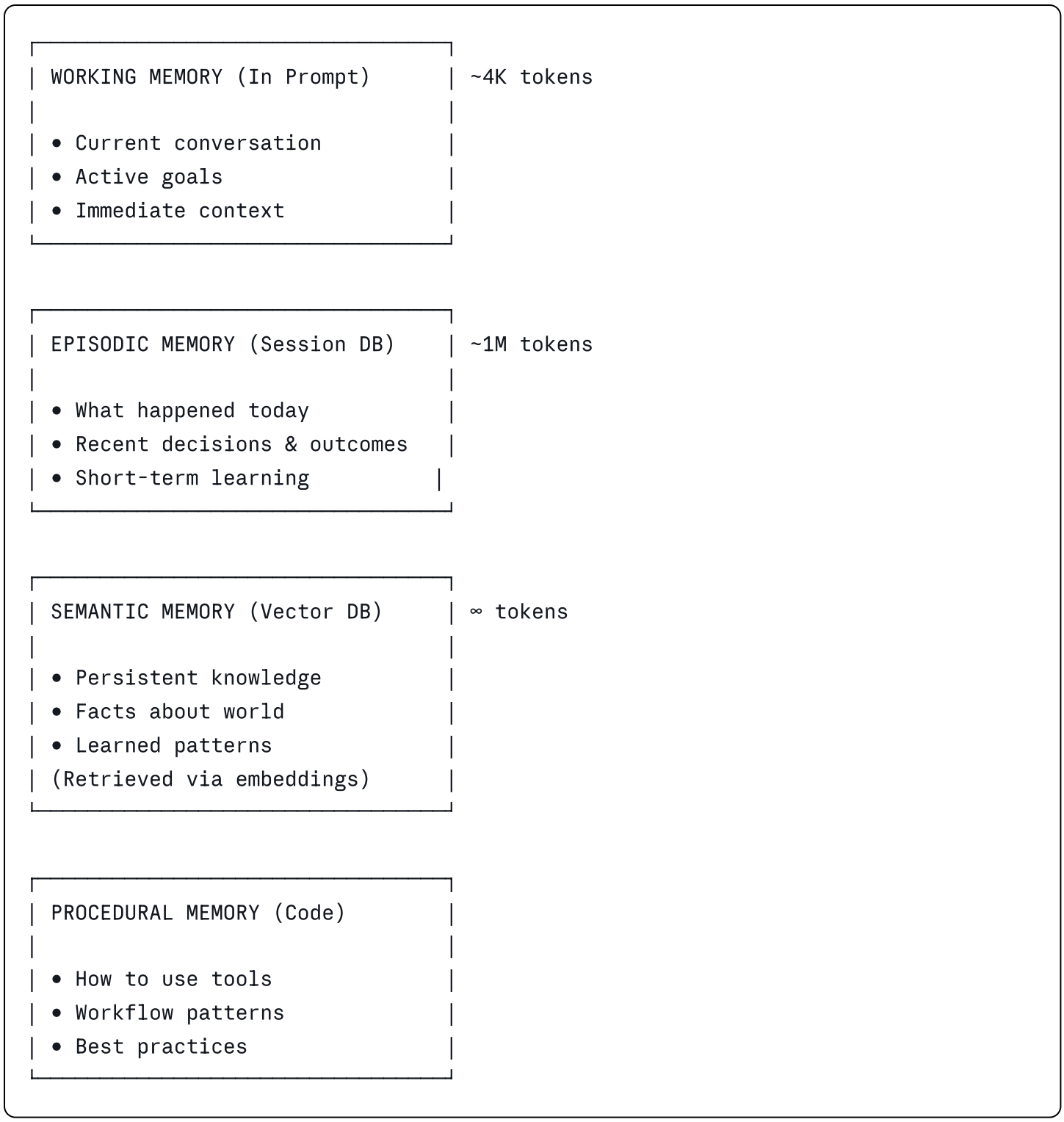
Parallel Execution Example

python

```python
import asyncio

async def parallel_agent(query: str):
    """
    Instead of sequential:
    task1 → task2 → task3

    Use parallel:
    task1 ─┐
    task2 ─┼─→ Combine results
    task3 ─┘
    """

    results = await asyncio.gather(
        execute_tool_async("search_option_a"),
        execute_tool_async("search_option_b"),
        execute_tool_async("search_option_c"),
    )

    # Combine results from all three searches
    return synthesize(results)
```

8.2 Memory Systems: Beyond Simple Context

Tiered Memory Architecture



Example: Memory Retrieval

python

```python
def agent_with_memory(query: str):
    # 1. Retrieve relevant episodic memory
    recent_context = memory_db.retrieve_recent(query, limit=3)

    # 2. Retrieve semantic memory (facts)
    facts = vector_db.search(embedding(query), top_k=5)

    # 3. Build prompt with limited context
    prompt = f"""
    Query: {query}

    Recent related conversations:
    {recent_context}

    Relevant facts:
    {facts}

    What should I do?
    """

    response = llm.complete(prompt)

    # 4. Store outcome in episodic memory
    memory_db.store(
        query=query,
        response=response,
        timestamp=now(),
        tags=["conversation_tag_1", "conversation_tag_2"]
    )

    return response
```

8.3 Tool Calling Strategies

Dynamic Tool Selection

Instead of defining all tools upfront, select based on context:

```python
```

```python
def smart_tool_selection(query: str, available_tools: List[Tool]):
    """
    Analyze query to determine which tools are needed.
    Don't send all tool definitions to LLM.
    """

    # Step 1: Classify the query
    query_type = llm.classify(query)
    # returns: "research", "transaction", "analysis", etc.

    # Step 2: Select tools for this query type
    needed_tools = tools_registry[query_type]

    # Step 3: Send only needed tools to LLM
    response = llm.complete(
        prompt=query,
        tools=needed_tools  # Much smaller than all tools!
    )

    return response
```

Tool Result Validation

python

```python
def validate_tool_result(tool_name: str, result: Any) -> bool:
    """
    Don't trust tool results. Validate them.
    """

    validators = {
        "calculate": lambda r: isinstance(r, (int, float)),
        "search": lambda r: isinstance(r, list) and len(r) > 0,
        "get_order": lambda r: "order_id" in r and "status" in r,
    }

    validator = validators.get(tool_name)
    if validator and not validator(result):
        return False  # Invalid result, ask agent to retry

    return True
```

8.4 Evaluation & Testing

Agent Quality Metrics

python

```python
@dataclass
class AgentMetrics:
    success_rate: float      # % of queries that complete successfully
    avg_iterations: float    # Average loops per query
    avg_tokens: float        # Average tokens per query
    cost_per_query: float    # Average cost
    latency_ms: float        # Response time
    hallucination_rate: float  # % of false claims
    user_satisfaction: float # 1-5 rating

def evaluate_agent(test_cases: List[TestCase]) -> AgentMetrics:
    """
    Test agent on benchmark dataset
    """

    results = []
    for test_case in test_cases:
        result = agent.process(test_case.query)

        results.append({
            'success': evaluate_correctness(result, test_case.expected),
            'iterations': test_case.iterations_used,
            'tokens': test_case.tokens_used,
            'cost': test_case.cost,
            'time': test_case.time_ms,
        })

    return AgentMetrics(
        success_rate=sum(r['success'] for r in results) / len(results),
        avg_iterations=sum(r['iterations'] for r in results) / len(results),
        # ... etc
    )
```

8.5 Agentic Patterns for LLMs at the Edge

For running agents on edge devices (phones, embedded):

python

```python
class EdgeAgent:
    """
    Lightweight agent for devices with limited compute.
    Uses smaller LLM, fewer tokens, simpler patterns.
    """

    def __init__(self):
        # Use smaller model
        self.llm = Ollama("mistral-7b")  # 7B params, ~4GB RAM

        # Limit context
        self.max_context_tokens = 2000

        # Use simpler patterns
        self.pattern = "loop"  # Not multi-agent

    def process(self, query: str) -> str:
        # Compress context aggressively
        context = self._compress_context(query)

        # Use quantized model
        with quantization.int4():
            response = self.llm.complete(context)

        return response
```

SECTION 9: REAL-WORLD USE CASES

Use Case 1: Customer Support Agent (Production)

Company: E-commerce platform

Problem: 10,000 support tickets/day, only 50 agents

Solution:

```
Customer Query
    ↓
[LOOP PATTERN AGENT]
├─ Search knowledge base
├─ Check order database
├─ Suggest solutions
└─ If >70% confident → respond
  Else → escalate to human

Results:
- Handles 70% of tickets automatically
- Reduces human agent load by 70%
- Cost: $0.15 per ticket
- Response time: 2 seconds
```

Use Case 2: Research Synthesis Agent (Academic)

Institution: Research university

Problem: Researchers drown in papers (500K daily in their field)

Solution:

```
[PLANNING PATTERN]
1. Search arXiv/PubMed for new papers
2. Summarize each paper (1 min read)
3. Identify relationships between papers
4. Synthesize into coherent insights
5. Generate weekly report

Saves each researcher: 10 hours/week
```

Use Case 3: Data Analyst Agent (Enterprise)

Company: Fortune 500 tech company

Problem: Business teams wait days for analytics reports

Solution:

```
[HIERARCHICAL PATTERN]
Manager Agent
  ├─ Worker: Data Extraction
  │  └─ Query databases
  ├─ Worker: Statistical Analysis
  │  └─ Run models
  ├─ Worker: Visualization
  │  └─ Create charts
  └─ Worker: Interpretation
     └─ Write insights

Results:
- Report generated in 2 minutes (vs. 2 days)
- 95% accuracy (humans verify 5%)
```

Use Case 4: Content Creation Agent (Media)

Company: News organization

Problem: Need to publish 200+ articles/day

Solution:

```
[REFLECTION PATTERN]
1. LLM drafts article
2. Fact-check against sources
3. If issues found → regenerate with feedback
4. Editor reviews final version

Results:
- Draft quality improves 3x faster
- Fact errors reduced from 5% to 0.1%
```

Use Case 5: Code Generation Agent (Development)

Company: Startups using AI pair programming

Solution:

```
[MULTI-AGENT PATTERN]
- Architect Agent: Design
- Coder Agent: Implementation
- Tester Agent: Validation
- Reviewer Agent: Code quality

Agents discuss, debate, reach consensus.
Result: Production-grade code suggestions.
```

SECTION 10: LEARNING PLAN (7-WEEK CURRICULUM)

WEEK 1: Foundations

Monday - Conceptual Foundations

- **Goal**: Understand what agents are
- **Reading**: Sections 1.1 - 1.4 of this document
- **Exercise**: Write down 3 examples of agents you use daily
- **Time**: 2-3 hours
- **Checklist**:
  - ☐ Know the observation-reasoning-action loop
  - ☐ Understand the difference between agents and traditional software
  - ☐ Know 4 types of memory

Tuesday - Historical Context & Reasoning

- **Goal**: Know how we got here
- **Reading**: Sections 1.3, 2.1 - 2.2
- **Exercise**: Draw the agent stack on paper
- **Time**: 2-3 hours
- **Checklist**:
  - ☐ Understand evolution from rules → ML → LLMs → agents
  - ☐ Know the transformer architecture basics
  - ☐ Understand token economy

Wednesday - State Machine & Token Budget

- **Goal**: Understand agent internals
- **Reading**: Sections 2.3 - 2.4
- **Exercise**: Calculate cost of a 5-iteration agent loop

- **Time:** 1-2 hours
- **Checklist:**
  - [ ] Know finite state machine states
  - [ ] Can estimate token usage
  - [ ] Know cost optimization strategies

Thursday - Hands-on: Run Existing Agents

- **Goal:** See agents in action
- **Task:**
  - Try OpenAI's ChatGPT with plugins
  - Try Claude with web search
  - Experiment with LangChain example
- **Time:** 2-3 hours
- **Reflection:** What did you notice? When did it fail?

Friday-Sunday - Review & Mini-Project

- **Goal:** Solidify Week 1 understanding
- **Project:** Build a simple chatbot that can:
  - Search Wikipedia
  - Answer basic questions
  - Knows when to ask for clarification
- **Time:** 3-4 hours
- **Deliverable:** Working code + README

WEEK 2: The Five Patterns (Part 1)

Monday - Loop Pattern

- **Goal:** Master the simplest pattern
- **Reading:** Section 3 - Pattern #1
- **Code:** Study the customer service agent example
- **Exercise:**
  - Implement Exercise 1 (simple loop agent)
  - Test with 5 different queries
  - Add 2 new tools

- **Time:** 3 hours
- **Checklist:**
  - [ ] Can implement loop pattern from scratch
  - [ ] Understand max iterations concept
  - [ ] Know when to use this pattern

Tuesday - Planning Pattern

- **Goal:** Build multi-step agents
- **Reading:** Section 3 - Pattern #2
- **Code:** Study recruitment agent example
- **Exercise:**
  - Implement planning pattern for trip planning
  - Create 3-step plan structure
  - Add tool calling between steps
- **Time:** 3-4 hours
- **Checklist:**
  - [ ] Understand plan → execute → synthesize flow
  - [ ] Can parse plan from LLM
  - [ ] Know pros/cons vs. Loop Pattern

Wednesday - Hierarchical Pattern

- **Goal:** Handle complex decomposition
- **Reading:** Section 3 - Pattern #3
- **Exercise:**
  - Design 3 worker agents for a problem
  - Implement basic manager-worker structure
  - Test communication between agents
- **Time:** 3-4 hours
- **Checklist:**
  - [ ] Understand decomposition concept
  - [ ] Can design worker agents
  - [ ] Know aggregation strategies

Thursday - Reflection & Multi-Agent Patterns

- **Goal:** Self-correction and collaboration
- **Reading:** Section 3 - Patterns #4 & #5

- **Exercise**:
  - Implement a simple reflection loop
  - Implement a 2-agent debate system
- **Time**: 3-4 hours
- **Checklist**:
  - [ ] Understand reflection as error correction
  - [ ] Know when multi-agent helps
  - [ ] Can design agent personas

Friday-Sunday - Pattern Integration Project

- **Project**: Build a task dispatcher that:
  - Analyzes query complexity
  - Chooses appropriate pattern (loop/planning/hierarchical)
  - Executes selected pattern
  - Returns result
- **Time**: 5-6 hours
- **Deliverable**: Pattern selector + 3 pattern implementations

---

WEEK 3: Architecture & Advanced Concepts

Monday - Agent Architecture Deep Dive

- **Goal**: Understand production systems
- **Reading**: Sections 2.2, 8.1
- **Exercise**: Design architecture for a real service
- **Time**: 2-3 hours

Tuesday - Memory Systems

- **Goal**: Build agents that remember
- **Reading**: Section 8.2
- **Code Exercise**:
  - Implement tiered memory (working + episodic + semantic)
  - Add vector database for similarity search
  - Test memory retrieval effectiveness
- **Time**: 3-4 hours

Wednesday - Tool Calling Strategies

- **Goal**: Optimize tool usage
- **Reading**: Section 8.3
- **Exercise**:
  - Design dynamic tool selection
  - Implement tool result validation
  - Build fallback tool paths
- **Time**: 3 hours

Thursday - Error Handling & Recovery

- **Goal**: Make robust agents
- **Exercise**: Implement Exercise 2 (error handling)
- **Build**: Comprehensive error handler for agent
- **Time**: 3-4 hours

Friday-Sunday - Memory + Architecture Project

- **Project**: Build a knowledge assistant that:
  - Remembers past conversations
  - Learns from interactions
  - Improves over time
  - Handles complex queries
- **Time**: 5-6 hours

WEEK 4: Testing, Evaluation & Production

Monday - Agent Evaluation Framework

- **Goal**: Measure agent quality
- **Reading**: Section 8.4
- **Exercise**:
  - Design test cases for an agent
  - Implement success metrics
  - Create evaluation suite
- **Time**: 2-3 hours

Tuesday - Cost Optimization

- **Goal**: Build production-efficient agents
- **Reading**: Section 2.4
- **Exercise**:
  - Profile token usage
  - Optimize prompts to reduce tokens
  - Compare costs of different patterns
- **Time**: 2-3 hours

Wednesday - Production Deployment

- **Goal**: Ship agents safely
- **Exercise**:
  - Add logging and monitoring
  - Implement rate limiting
  - Create fallback handlers
  - Build admin dashboard
- **Time**: 3-4 hours

Thursday - Security & Safety

- **Goal**: Prevent failures
- **Topics**:
  - Input validation
  - Tool access control
  - Output filtering
  - Audit logging
- **Time**: 2-3 hours

Friday-Sunday - Evaluation & Production Project

- **Project**: Build complete agent system with:
  - Comprehensive testing suite
  - Cost tracking
  - Production-ready deployment
  - Monitoring dashboard
  - A/B testing capability

- **Time:** 6-8 hours

WEEK 5: Mini-Projects (Intermediate Level)

Complete Exercise 3 (weather recommendation) and Exercise 4 (self-correcting math agent).

Tuesday - Project 1: Travel Planning Agent

- **Requirements:** Section 6, Exercise 5
- **Time:** 6-8 hours
- **Deliverable:** End-to-end agent with multiple patterns

Wednesday - Project 2: Technical Support Bot

- **Build:** Support bot for your product
- **Features:**
  - Knowledge base search
  - Ticket creation
  - Escalation
  - Self-learning from feedback
- **Time:** 6-8 hours

Thursday-Friday - Code Review & Optimization

- **Review:** Check your projects for:
  - Code quality
  - Error handling
  - Token efficiency
  - Test coverage
- **Optimize:** Improve weakest aspects

Weekend - Reflection & Documentation

- **Document:** Your learning
- **Write:** Summary of 5 patterns
- **Reflect:** Which patterns work best for what?

WEEK 6: Real-World Applications

Focus: Learning from production systems

Monday - Analyze Existing Agents

- **Task**: Study these open-source agents:
  - AutoGPT
  - Baby AGI
  - LangChain examples
  - LlamaIndex implementations
- **Document**: Patterns you identify
- **Time**: 3-4 hours

Tuesday-Wednesday - Case Study Deep Dives

- **Read**: Use Case examples (Section 9)
- **Analyze**: 5 different real-world implementations
- **Extract**: Lessons learned
- **Time**: 4-5 hours

Thursday - Building Your Own Agent Framework

- **Create**: Simple but elegant agent framework
- **Features**:
  - Tool registry
  - Memory management
  - Error handling
  - Logging
  - Extensibility
- **Time**: 4-5 hours

Friday-Sunday - Build Production Agent

- **Project**: Create agent for real use case
- **Must include**:
  - Proper error handling
  - Cost monitoring
  - Performance metrics
  - User feedback loop

    - Documentation
- **Time**: 8-10 hours

WEEK 7: Mastery & Specialization

Focus: Deep expertise in your area of interest

Monday - Choose Specialization

Select ONE:

- **A)** Financial analysis agents
- **B)** Content generation agents
- **C)** Research & synthesis agents
- **D)** Code generation agents
- **E)** Data processing agents

Tuesday-Thursday - Deep Specialization

- **Read**: Advanced materials in your domain
- **Build**: Specialized agent system
- **Optimize**: For your specific use case
- **Test**: Extensively
- **Time**: 12-15 hours

Friday - Capstone Project Planning

- **Design**: Ambitious agent project
- **Scope**: 2-4 weeks of work
- **Details**: Architecture, patterns, tools, evaluation

Saturday-Sunday - Capstone Kickoff

- **Build**: First version of capstone
- **Goal**: Demonstrate mastery of 3+ patterns
- **Polish**: Code, documentation, testing
- **Time**: 8-10 hours

After Week 7: Capstone Project (Self-Paced)

Goal: Build something impressive and useful

Examples:

1. **Research Assistant Agent**: Summarizes papers, finds connections, generates insights
2. **Code Reviewer Agent**: Reviews code, suggests improvements, explains changes
3. **Content Editor Agent**: Writes, fact-checks, edits, publishes
4. **Business Analyst Agent**: Analyzes data, creates reports, provides recommendations
5. **Custom Agent**: For your specific domain/problem

Requirements:

- Uses 2+ agentic patterns
- Production-ready code
- Comprehensive testing
- Documentation
- Demo/presentation

SECTION 11: TOOLS & FRAMEWORKS (2026 Edition)

11.1 LLM Providers (For Reasoning)

Provider	Best For	Models	Cost
OpenAI	General use, function calling	GPT-4, GPT-4 turbo	$0.5-15/1M tokens
Anthropic	Long context, safety	Claude Opus, Sonnet	$3-15/1M tokens
Google	Multimodal, embedding	Gemini 2.0, PaLM	$0.075-2/1M tokens
Open Source	Local deployment, privacy	Llama 2, Mistral, Qwen	Free (self-hosted)
Specialized	Domain-specific	MedLM, FinGPT, etc.	Custom pricing

11.2 Agent Frameworks (Orchestration)

LangChain (Most Popular)

```python
```

```python
from langchain.agents import initialize_agent, load_tools
from langchain.llms import OpenAI

llm = OpenAI(api_key="...")
tools = load_tools(["serpapi", "llm-math"])
agent = initialize_agent(tools, llm, agent="zero-shot-react-description")
agent.run("What's 2+3?")
```

Strengths: Large ecosystem, many integrations, good documentation **Weaknesses:** Large overhead, sometimes slow

LlamaIndex (For Data)

```python
from llama_index import GPTVectorStoreIndex, Document

documents = [Document(text="...")]
index = GPTVectorStoreIndex.from_documents(documents)
response = index.query("What is X?")
```

Strengths: Optimized for RAG, simple API **Weaknesses:** Limited to data retrieval

AutoGPT (Full Agent Framework)

```python
# Understands complex goals
agent = AutoGPT(goal="Find best GPU for ML, <$2000")
# Automatically uses web search, shopping APIs, etc.
result = agent.execute()
```

Strengths: Autonomous goal achievement, general-purpose **Weaknesses:** Can be unpredictable, costs money

Crew (Multi-Agent Orchestration)

```python
```

```python
from crewai import Agent, Task, Crew

agents = [
    Agent(role="Researcher", goal="Find information"),
    Agent(role="Writer", goal="Create content"),
    Agent(role="Editor", goal="Polish content"),
]

crew = Crew(agents=agents, tasks=[...])
result = crew.kickoff()
```

Strengths: Elegant multi-agent, good defaults **Weaknesses**: Newer, smaller community

11.3 Vector Databases (For Memory)

Database	Best For	Features
Pinecone	Managed vector DB	Serverless, auto-scaling
Weaviate	Open-source vector DB	Fast, modular
Qdrant	High-performance	GPU optimization
Milvus	Enterprise-scale	Distributed
Pgvector	PostgreSQL extension	Using existing DB

11.4 API / Tool Calling Services

Service	Capability	Use Case
Langfuse	LLM tracing & debugging	Monitor agent calls
Humanloop	Prompt management	Version control prompts
Weights & Biases	Agent monitoring	Track performance metrics
Replit Agent	Sandboxed code execution	Safe tool calling
Zapier API	7000+ integrations	Connect to existing tools

11.5 Recommended Tech Stack (2026)

```
For Building Agents:
├─ Core Language
|  └─ Python 3.11+ (best support for AI)
```

```
|
├─ LLM Selection
|  └─ OpenAI GPT-4 (best general performance)
|     Or Claude (better for reasoning)
|     Or Llama (for local/cost savings)
|
├─ Agent Framework
|  └─ LangChain (most mature)
|     Or LlamaIndex (if data-heavy)
|     Or Anthropic SDK (if Claude-focused)
|
├─ Memory / RAG
|  └─ Pinecone (simplest)
|     Or Weaviate (self-hosted)
|
├─ Monitoring
|  └─ Langfuse (best observability)
|
├─ Deployment
|  └─ FastAPI (for API serving)
|     Or Vercel (for serverless)
|     Or Modal (for AI workloads)
|
└─ Infrastructure
   └─ Docker (containerization)
      Kubernetes (scaling)
```

SECTION 12: FURTHER LEARNING RESOURCES

12.1 Essential Reading

Books

1. **"Building AI Applications with Python"** - Lex Fridman, Andrew Ng (2025)
   - Comprehensive guide to building production agents
   - Code examples in each chapter
2. **"Designing Autonomous Agents"** - Russell & Norvig (latest edition)
   - Academic foundations
   - Theoretical understanding
3. **"LLM Engineering in Production"** - OpenAI Team (2024)
   - Best practices from production systems
   - Real-world challenges and solutions

Papers (ArXiv)

- "Agent Design Patterns for LLMs" - Anthropic Team, 2024
- "Multi-Agent Systems: A Survey" - Shoham & Leyton-Brown, 2023
- "In-Context Learning for Agents" - Google DeepMind, 2024
- "Agentic Reasoning with Language Models" - OpenAI, 2024

Websites & Blogs

- **Full-Stack Python**: fullstackpython.com/ai-agents
- **Hugging Face Blog**: huggingface.co/blog/agents
- **Anthropic Blog**: anthropic.com/blog
- **OpenAI Documentation**: platform.openai.com/docs

12.2 Communities & Support

- **GitHub Discussions**: Star and follow LangChain, LlamaIndex
- **Discord Communities**:
  - LangChain Discord (10K+ members)
  - Anthropic Discord
  - AI Engineers Community
- **Reddit**: r/LocalLLaMA, r/LanguageModels, r/OpenAI
- **Conferences**: AI Summit 2026, NeurIPS, ICLR

12.3 Tools to Explore

1. **OpenAI Playground**: Test agents interactively
2. **LangChain Hub**: Pre-built agents and prompts
3. **Llama Playground**: Run local models easily
4. **AnthropicLabs**: Experimental features
5. **GitHub Copilot for Agents**: IDE integration

12.4 Specialized Paths

If interested in Data Agents:

- Learn SQL optimization
- Study vector databases
- Master prompt engineering for queries
- Tools: LlamaIndex, Langchain-SQL

If interested in Coding Agents:

- Learn abstract syntax trees (ASTs)
- Study code analysis
- Master debugging strategies
- Tools: Replit Agent, Github Copilot

If interested in Multi-Agent Systems:

- Study game theory
- Learn consensus algorithms
- Master agent communication protocols
- Tools: Crew, AutoGPT, Hugging Face Smol

If interested in Edge Agents:

- Study model quantization
- Learn inference optimization
- Study mobile ML
- Tools: ONNX, CoreML, TensorFlow Lite

12.5 Staying Current (2026+)

LLMs and agents change rapidly. Stay updated:

Weekly:

- Follow r/LocalLLaMA on Reddit
- Subscribe to "The Batch" (Andrew Ng)
- Check Hugging Face Weekly Digest

Monthly:

- Read ArXiv papers (use arxiv-sanity.com)
- Attend relevant meetups
- Try new frameworks

Quarterly:

- Take courses on Coursera/Fast.ai
- Review benchmarks (LLMQA, LMM-Eval)
- Build something new to stay hands-on

CONCLUSION

What You Now Know

You've learned:

1. ✅ What agentic systems are and why they matter
2. ✅ 5 major design patterns (Loop, Planning, Hierarchical, Reflection, Multi-Agent)
3. ✅ Deep technical understanding of how agents work
4. ✅ How to build and deploy production-ready agents
5. ✅ Common mistakes and how to avoid them
6. ✅ Advanced concepts for optimization
7. ✅ A structured 7-week learning plan
8. ✅ Tools and frameworks to use

Your Next Steps

1. **This Week:** Complete Week 1 of the learning plan
2. **This Month:** Build 2-3 working agent projects
3. **This Quarter:** Deploy an agent to production
4. **This Year:** Become a recognized expert in agentic systems

Remember

> "The future of software isn't code. It's agents that understand goals and figure out how to achieve them."
> — The future is collaborative between humans and AI agents working together.

APPENDIX: Quick Reference

Pattern Decision Matrix

```
Problem Type                → Best Pattern
──────────────────────────────────────────
Answer a question           → Loop
Multi-step task             → Planning
Very complex problem        → Hierarchical
Uncertain success           → Reflection
Need multiple viewpoints    → Multi-Agent
```

Cost Quick Estimation

```
Simple query (Loop, 1 iteration):
  Input: 2000 tokens, Output: 200 tokens
  Cost: $0.01-0.05

Complex query (Planning, 5 iterations):
  Input: 10000 tokens, Output: 1000 tokens
  Cost: $0.15-0.50

Very complex (Hierarchical, 10 worker calls):
  Input: 50000 tokens, Output: 5000 tokens
  Cost: $0.75-2.50
```

Token Budgets (Approximate)

```
System Prompt:      300-500 tokens
Tool Definitions:   800-1500 tokens
Few-shot Examples:  500-2000 tokens
Context/History:    2000-10000 tokens
Current Query:      50-200 tokens
Safe Max:           Should stay under 80% of limit
```

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END OF DOCUMENT

Quick Navigation Guide

- **For Beginners:** Start with Section 1, then Section 3 (first 2 patterns)
- **For Intermediates:** Study Sections 2-3, then Section 5
- **For Advanced:** Focus on Sections 8-10
- **For Hands-On:** Jump to Section 6 (Exercises)
- **For Reference:** Use Appendix for quick lookups
- **For Implementation:** Use Section 5 (Code Examples) and Section 11 (Tools)